Ans : (d)

Average number $= \frac{2000+1000+3500+5500}{4}$

$= \frac{12000}{4} = 3000$

Direction (71 to 73)

71. The given table represents the marks obtained by four students W, X, Y and Z in four subjects P, C, B and M with the maximum marks in each subject being 100.

The average marks of all the four students in subject C is.

Students/ Subject	P	C	B	M
W	70	90	50	85
X	55	80	95	60
Y	60	20	90	40
Z	90	80	40	65

(a) 67.5 (b) 67

(c) 67.75 (d) 67.25

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (a)

Average marks of all four students in subject C

$= \frac{90+80+20+80}{4} = \frac{270}{4} = 67.5$

72. What is the average mark of four students in M?

(a) 62 (b) 62.25

(c) 62.75 (d) 62.5

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (d) Average marks of all four students in M

$= \frac{85+60+40+65}{4}$

$= \frac{250}{4} = 62.5$

73. What is the average mark of the four students in P?

(a) 68.5 (b) 68

(c) 68.75 (d) 68.25

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (c) Average marks of all four students in P

$= \frac{70+55+60+90}{4} = \frac{275}{4} = 68.75$

Type - 8

74. A person expenditure has increased ₹10000 in each month in February and March. If his expenditure in January was ₹10000, then find the average expenditure (in ₹) from January to march.

(a) 20000 (b) 15000

(c) 10000 (d) 25000

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) January expenses of the person = ₹10000

expenses for the month of february

= 10000 + 10000 = ₹20000

And expenses for the month of March

= 20000 + 10000 = ₹30000

Average expenses from January to March

$= \frac{10000+20000+30000}{3}$

$= \frac{60000}{3}$ = ₹20000

75. A person expenditure has increased by ₹50000 in the months of February and March. If his expenditure in January was ₹50000, then find his average expenditure (in ₹) from January to March.

(a) 100000 (b) 150000

(c) 75000 (d) 50000

RRB NTPC 27.04.2016 Shift : 3

Ans : (a) Expense in January Month = ₹ 50000

∵ In the month of February and March, expenses increase by ₹ 50000

∴ Expenses for the month of February

= 50000 + 50000

= ₹ 100000

Expenses for the month of March = 100000 + 50000

= ₹ 150000

Hence, Average expenses from January to March

$= \frac{50000+100000+150000}{3}$

$= \frac{300000}{3}$ = ₹ 100000

18.

Speed, Time & Distance

Type - 8

1. Two buses from a house run at a speed of 25 km/h at an interval of 15 minutes. How much more speed (km/h) does a woman coming from the opposite side of the house have to walk so that the buses meet at an interval of 10 minutes.

(a) 12 (b) 12.25
(c) 12.5 (d) 12.75

RRB RPF SI – 06/01/2019 (Shift-III)

Ans : (c) Speed of bus = 25 Km./hr.

Let the speed of woman = x km/h

Distance = D, Time = 15 minute $= \frac{15}{60} = \frac{1}{4}$ hours

then time after women's speed = 10 minute

$= \frac{10}{60} = \frac{1}{6}$ hours

Then Relative speed $(S) = \frac{D}{T}$

$\Rightarrow \quad D = S \times T$

$D = 25 \times \frac{1}{4}$

$\because \quad D = \frac{25}{4}$ .....(i)

$D = \frac{25+x}{6}$ .....(ii)

from equation (i) and equation (ii)

$\frac{25+x}{6} = \frac{25}{4}$

$25 + x = \frac{150}{4}$

$x = \frac{150}{4} - 25$

$x = \frac{150-100}{4}$

$x = \frac{50}{4}$

Speed of woman x = 12.5 Km./hr.

2. A car covers a certain distance at a speed of 50 km/h in 8 hours. How much should its speed increase to cover the same distance in 5 hours?

(a) 50 Km./hr. (b) 40 Km./hr.
(c) 30 Km./hr. (d) 80 Km./hr.

RRB RPF Constable – 19/01/2019 (Shift-III)

Ans : (c) Let distance = x km

According to question,

$\text{speed} = \frac{\text{distance}}{\text{time}}$

$50 = \frac{x}{8}$

$\boxed{x = 400\text{km}}$

After covering 400 km distance in 5 hours

$\text{speed} = \frac{400}{5} = 80\text{km/h}$

then,

speed increase = 80–50= 30Km./hr.

3. The speed of 14 meters per second is equal to:

(a) 46.6 Km./hr. (b) 50.4 Km./hr.
(c) 70 Km./hr. (d) 28 Km./hr.

RRB RPF SI – 05/01/2019 (Shift-I)

Ans : (b) The speed of 14 meter per second is the same as 50.4 Km./hr.

1 m/sec = 18/5 Km./hr.

14 m/sec = $\frac{18}{5} \times 14 =$ 50.4 Km./hr.

4. Prithi is going to Delhi from Rajdhani express which is running 6 minutes late. Driver in creases. It's speed by 4 km/hr. By doing this, the train arrives on time at the next station which is at the distance of 36 km. Find the actual speed of the train.

(a) 20 Km./hr. (b) 26 Km./hr.
(c) 36 Km./hr. (d) 30 Km./hr.

RRB RPF Constable – 17/01/2019 (Shift-I)

Ans. (c) : Let the speed of train = x Km./hr.

Distance between both stations = 36 Km.

According to question,

$\frac{36}{x+4} = \frac{36}{x} - \frac{6}{60}$

(time = distance/speed)

$\Rightarrow 36\left(\frac{1}{x} - \frac{1}{x+4}\right) = \frac{1}{10}$

$\Rightarrow 36\left(\frac{x+4-x}{x(4+x)}\right) = \frac{1}{10}$

$\Rightarrow 36 \times 4 \times 10 = x^2 + 4x$

$\Rightarrow x^2 + 4x - 1440 = 0$

$\Rightarrow x^2 + 40x - 36x - 1440 = 0$

$\Rightarrow x(x+40) - 36(x+40) = 0$

$\Rightarrow (x+40)(x-36) = 0$

$\therefore x = 36\text{Km/h}$

5. Kishan cycled 96 km at a certain speed. If he cycled 4 km/h slower, then he would have taken an addition time of two hours to reach the destination. What is the speed, at which kishan actually cycled in Kilometers?

(a) 12 (b) 18
(c) 16 (d) 15

RRB Group-D–11/10/2018 (Shift-II)

Ans : (c) Let actual speed = x Km./hr.
Distance = 96 km
New speed = (x–4) Km./hr.
According to question,

$$\frac{96}{x-4}-\frac{96}{x}=2$$

$$\frac{48}{x-4}-\frac{48}{x}=1$$

$$48\left(\frac{1}{x-4}-\frac{1}{x}\right)=1$$

$$48\left(\frac{x-x+4}{x(x-4)}\right)=1$$

$48 \times 4 = x(x-4)$
$x^2 - 4x - 192 = 0$
$x^2 - 16x + 12x - 192 = 0$
$x(x-16) + 12(x-16) = 0$
$(x+12)(x-16) = 0$
Hence $x - 16 = 0$
$x = 16 \quad x \neq -12$
Actual speed = 16 Km./hr.

6. A person has to cover a distance of 40 km. He covers a distance of 16 km and the remaining distance is covered by a tanga. If he covers a distance of 16 km. from a tanga and the remaining distance at a speed of 4 km/hr. So takes more than 1 hour. Find the speed of the tanga.

(a) 12 Km./hr. (b) 8 Km./hr.
(c) 16 Km./hr. (d) 10 Km./hr.

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (b) Let the speed of taanga = x Km./hr.
Distance = 40 km
According to question,

$$\left(\frac{16}{x}+6\right)-\left(\frac{24}{x}+4\right)=1$$

$$\Rightarrow \frac{16}{x}+6-\frac{24}{x}-4=1$$

$$\Rightarrow \frac{16}{x}-\frac{24}{x}=1-2$$

$$\Rightarrow \frac{16-24}{x}=-1$$

$$\Rightarrow -8=-x$$

x = 8 Km./hr.

7. In a 200 meters long race the runner A beats runner B by 3 seconds. If the speed difference between A and B is 1.5 m/s. So find the speed of A in meter per second.

(a) 10.778 (b) 10.5
(c) 8.728 (d) 9.728

RRB Group-D – 15/11/2018 (Shift-III)

Ans. : (a) Let the speed of B = x m/s
$\therefore$ Speed of A = (x + 1.5) m/s
According to question,

$$\frac{200}{x}-\frac{200}{(x+1.5)}=3$$

$$\frac{200x+300-200x}{x^2+1.5x}=3$$

$x^2 + 1.5x - 100 = 0$
$x^2 + 10.778x - 9.278x - 100 = 0$
$x(x + 10.778) - 9.278(x + 10.778) = 0$
$(x - 9.278)(x + 10.778) = 0$
x = 9.278 m/s
Hence, Speed of A = (9.278 + 1.5) = 10.778 m/s

8. A car in a fog crosses a person, who was walking in the same direction at a speed of 3 km/h. That person could see this car for 100 m up to 4 minutes. What was the speed of the car?

(a) 9/2 Km./hr. (b) 7/2 Km./hr.
(c) 5 Km./hr. (d) 5/2 Km./hr.

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (a) : Let the speed of train = x Km./hr.
According to question,

$$\left\{\text{Relative speed}=\frac{\text{Distance}}{\text{Time}}\right\}$$

$$\because \quad x-3=\frac{\frac{100}{1000}}{\frac{4}{60}} \qquad \left\{\because 100\,\text{m.}=\frac{100}{1000}\text{km and } 4\,\text{minute}=\frac{4}{60}\text{hr}\right\}$$

$$x-3=\frac{15}{10}$$

$$x=\frac{3}{2}+3$$

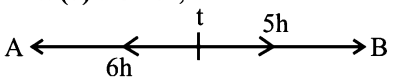
$$x=\frac{9}{2}\text{km/hr}$$

9. Two cars A and B starting at the same time meet each other in opposite direction after t hours and after arriving they reach their destination after 5 hours and 6 hours. If the speed of car A is 55 km/hr, what will be the speed of the car B?

(a) $66\sqrt{12}$ km/hr (b) $110\sqrt{3}$ km/hr
(c) $\frac{110}{\sqrt{6}}$ km/hr (d) $\frac{55}{6}\sqrt{30}$ km/hr

RRB Group-D – 05/11/2018 (Shift-I)

Ans. (d) : Given,



Time taken to reach destination after meeting (A) = 5 hr.
Time taken to reach destination after meeting (B) = 6 hr.
Speed of A = 55 Km./hr.
Let Speed of B = x Km./hr.

$\therefore \frac{S_B}{S_A} = \sqrt{\frac{t_A}{t_B}}$

$\frac{x}{55} = \sqrt{\frac{5}{6}}$

$x = 55 \times \sqrt{\frac{5}{6}}$

$x = 55 \times \sqrt{\frac{5 \times 6}{6 \times 6}}$

$x = \frac{55}{6} \times \sqrt{30}$

$x = \frac{55}{6}\sqrt{30}$ km/hr

10. Parvej belongs to town A and Gautam belongs to Town B. They start their journey to each other cities by the same route at the same time. They meet some where along the way and continue their journey. Parvej takes 12 hours to reach the destination after meeting Gautam, while Gautam takes another 3 hours to reach the city of parvej. If parvej travels at a speed of 60 km/hr, then find the speed of Gautam in km/hr.

(a) 120 (b) 105
(c) 90 (d) 125

RRB Group-D – 11/12/2018 (Shift-I)

Ans. (a) :

$\frac{v_1}{v_2} = \sqrt{\frac{t_2}{t_1}}$

$\frac{60}{v_2} = \sqrt{\frac{3}{12}}$ $\{\because v_1 = 60$ Km./hr., $t_1 = 12, t_2 = 3\}$

$\frac{60}{v_2} = \sqrt{\frac{1}{4}}$

$v_2 = 120$ Km./hr.

11. Two people are cycling in the same direction a train traveling at a fixed speed passes from there. The train crosses the first in 8 seconds and the second in 8.4 seconds, the first person was cycling at a speed of 9 km/hr while the second person was cycling at a speed of 12 km/hr. What is the speed of the train in km/hr.

(a) 66 (b) 75
(c) 81 (d) 72

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (d) Let the speed of train is V Km./hr.
Train is going in the direction of both the people.
From rule of relative speed, distance = speed × time

(for the first person) distance $= (V-9) \times \frac{8}{3600}$

(for the second person) distance $= (V-12) \times \frac{8.4}{3600}$

$\because$ The distance travell by the train crossing both person will be same.

$\therefore (V-9) \times \frac{8}{3600} = (V-12) \times \frac{8.4}{3600}$

$8V - 72 = 8.4V - 100.8$

$100.8 - 72 = 0.4V$

$28.8 = 0.4V$

$V = \frac{288}{4} = 72 km/h$

12. Two cars travel from the same houses at a speed of 20 km/h at an interval of 10 minutes. At what speed does a woman come in the opposite direction towards the house. If she gets cars in 8 minutes interval.

(a) 5 (b) 6
(c) 7 (d) 4

RRB NTPC 17.01.2017 Shift-1

Ans : (a) Let the speed of woman is x Km./hr. and the woman's distance from the house is l km and at t time the woman gets the first car.
Then-

$x \times t + 20t = l$ …….. (i)

and $\left(t + \frac{8}{60}\right)x + \left(t + \frac{8-10}{60}\right) \times 20 = l$

$xt + \frac{8x}{60} + 20t - \frac{20 \times 2}{60} = xt + 20t$

$\frac{8x}{60} = \frac{40}{60}$

x = 5 Km./hr.

13. The speed of the car A is 80 km/h. If it covers a certain distance in 6 hours and B travels 60 km less than the car A in that time. What is the speed of car B?

(a) 50 Km./hr. (b) 70 Km./hr.
(c) 60 Km./hr. (d) 55 Km./hr.

RRB NTPC 31.03.2016 Shift : 1

Ans : (b) Distance traveled by car A = 80×6 = 480km.
Distance travelled by car B = 480 – 60 = 420 km.

$\therefore$ Speed of car B $= \frac{420}{6} = 70$ Km./hr.

14. The ratio between the speeds of two buses is 11:9. If the second bus runs 270 kms in 15 hours, then the speed of the first bus is?

(a) 23 Km./hr. (b) 11 Km./hr.
(c) 2 Km./hr. (d) 22 Km./hr.

RRB NTPC 18.04.2016 Shift : 1

Ans : (d) Given-
Speed of two buses is 11 x and 9x respectively.

Speed of second bus $= \frac{270}{15} = 18\,km/h.$

$\Rightarrow 9x = 18$ Km./hr.

$\Rightarrow x = 18/9$

$\Rightarrow x = 2$ Km./hr.

Hence speed of first bus $= 11x = 11\times2 = 22$ Km./hr.

15. Two bicycles from a house started at a speed of 24 km/h with a interval of 15 minutes. How much more speed does a woman coming from the opposite direction of the house have to walk so that she meet a cycle at an interval of 10 minutes.

(a) 13 (b) 11
(c) 12 (d) 14

RRB NTPC 11.04.2016 Shift : 1

Ans : (c) Distance travelled by cycle in 15 minutes

$$= 24\times\frac{15}{60} = 6 \text{ km.}$$

Hence speed of women $= \frac{6}{\frac{10}{60}} = \frac{6\times60}{10} = 36$ Km./hr.

$\therefore$ Intended speed $= 36–24 = 12$ Km./hr.

16. Two jeep runs from a house at a speed of 20 km/h at 10 minutes interval. How much speed will a woman coming from the opposite direction of the house have to walk so that she will meet the second jeep at a interval of 6 minutes.

(a) 13 (b) $13\frac{2}{3}$
(c) $13\frac{1}{3}$ (d) 13.5

RRB NTPC 11.04.2016 Shift : 3

Ans : (c) Let the speed of woman = x Km./hr.
According to question,

$$x\times\frac{6}{60} = 20\times\frac{4}{60}$$

$$x = \frac{40}{3} = 13\frac{1}{3} \text{ Km./hr.}$$

17. If Vinay would have run at a high speed of 2 km/h he would have taken 10 minutes less to cover 4 km, then find the speed of vinay.

(a) 7 Km./hr. (b) 5 Km./hr.
(c) 4 Km./hr. (d) 6 Km./hr.

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans : (d) Let the speed from starting = S Km./hr.
Increased speed = (S + 2) Km./hr.
Distance = 4 km
from use of formula -

$$\frac{\text{Product of speed}}{\text{Distance}} = \frac{\text{difference in speed}}{\text{difference in time}}$$

$$\frac{S(S+2)}{4} = \frac{2}{\frac{10}{60}}$$

$$\frac{S^2+2S}{4} = 12$$

$$S^2+2S-48=0$$

$$S^2+8S-6S-48=0$$

$$S(S+8)-6(S+8)=0$$

$$(S+8)(S-6)=0$$

$$S+8=0 \Rightarrow S=-8$$

$$S-6=0 \Rightarrow S=6\,km/hr$$

Since the speed is always positive therefore intended answer = 6 Km./hr.

18. Two mini buses run from a house at an interval of 10 minutes at a speed of 22 km/h. How much more speed (km/h) a person coming from the opposite direction of the house will have to walk so that they can settled at an interval of 6 minutes.

(a) 14 (b) $14\frac{1}{3}$ (c) $14\frac{2}{3}$ (d) 14.5

RRB NTPC 07.04.2016 Shift : 1

Ans : (c) Let the speed of person = x Km./hr.
According to question,

$$22+x = \frac{22\times10}{6} = 22+x = \frac{220}{6}$$

$$x = \frac{110}{3} - 22$$

$$x = \frac{110-66}{3} = \frac{44}{3}$$

$$x = 14\frac{2}{3} km/h$$

19. Two bicycles start from a house together at an interval of 10 minutes and run at a speed of 6 km/hr. What should be the speed of a woman coming from the opposite direction towards home so that she can cross both the bicycles at an interval of 8 minutes.

(a) 1.7 (b) 1.6
(c) 1.5 (d) 1.4

RRB NTPC 19.01.2017 Shift : 1

Ans : (c)
Distance covered by bicycle in 10 minutes

$$= 6\times\frac{10}{60} = 1km.$$

Let the speed of woman = x Km./hr.
Then Relative speed = (6+x) Km./hr.

$$6+x = \frac{1}{\frac{8}{60}} = \frac{60}{8} \Rightarrow x = \frac{60}{8} - 6$$

$$x = \frac{60-48}{8} = \frac{12}{8}$$

$$x = \frac{3}{2} = 1.5 km/h$$

20. A car covers a distance of 240 km in 6 hours. How much should the speed of the car be increased to cover 1½ of a distance?

(a) 15 Km./hr. (b) 20 Km./hr.
(c) 25 Km./hr. (d) 30 Km./hr.

RRB NTPC 26.04.2016 Shift : 1

Ans : (b) Speed $= \frac{\text{Distance}}{\text{Time}} = \frac{240}{6} = 40$ Km./hr.

New Distance = $240 \times 1\frac{1}{2} = 240 \times \frac{3}{2} = 360$ km.

Again Speed $= \frac{\text{distance}}{\text{time}} = \frac{360}{6} = 60$ Km./hr.

Speed Increase = 60 – 40 = 20 Km./hr.

21. Two vehicles from a house moved at a speed of 25 km/h. in a interval of 20 minutes. How much more speed a woman coming from the opposite direction of the house will have to walk so that she gets a vehicle at an interval of 18 minutes.

(a) 2 (b) $2\frac{5}{9}$
(c) $2\frac{7}{9}$ (d) $2\frac{8}{9}$

RRB NTPC 26.04.2016 Shift : 2

Ans : (c) Distance covered by vehicle in 20 minutes

Distance = Speed × Time

$= 25 \times \frac{20}{60}$ km.

$= 25 \times \frac{1}{3} = \frac{25}{3}$ km.

Let the speed of women = x Km./hr.

∴ From Question,

$$\frac{\frac{25}{3}}{25+x} = \frac{18}{60}$$

$$\Rightarrow \frac{25}{3(25+x)} = \frac{18}{60}$$

$$\Rightarrow \frac{25}{75+3x} = \frac{18}{60}$$

$$\Rightarrow \frac{25}{75+3x} = \frac{3}{10}$$

$$\Rightarrow 250 - 225 = 9x$$

$$\Rightarrow 25 = 9x$$

$$\Rightarrow x = \frac{25}{9}$$

Hence speed of women = $2\frac{7}{9}$ Km./hr.

22. Two van from one house ran at a speed of 20 km/h at an interval of 16 minutes. What more speed a woman coming from the opposite direction of the house will have to walk go get the van at an interval of 12 minutes.

(a) 6 (b) $6\frac{1}{3}$
(c) $6\frac{2}{3}$ (d) 6.5

RRB NTPC 26.04.2016 Shift : 3

Ans : (c) Let the women walk x Km./hr. more,

from formula- Distance = speed × time

$$\frac{16}{60} \times 20 = (x+20) \times \frac{12}{60}$$

$$320 = 12x + 240$$

$$12x = 80$$

$$x = \frac{80}{12} = \frac{20}{3} = 6\frac{2}{3} \text{km/hr.}$$

23. A Thief drives his motor cycle at a speed of x km/h. 4 hours later a policeman starts chasing him in a jeep and captures him in 4 hours. If the average speed of van of the policeman is 40 km/hr then find the speed of the thief's motorcycle in km/hr.

(a) 30 (b) 36
(c) 45 (d) 20

RRB NTPC 27.04.2016 Shift : 1

Ans : (d) The policeman catches the thief, so the distance covered by both will be the same.

x × 8 = 40 × 4

x = 20

∴ Speed of motorcycle of thief = 20 Km./hr.

24. A thief drives his motorcycle at a speed of x km/h. A police started chasing him with jeep after 6 hours and caught him after 4 hours. If the average speed of a police is 90 km/h, then find the speed of the thief's motorcycle in km/hr.

(a) 30 (b) 36
(c) 45 (d) 60

RRB NTPC 30.04.2016 Shift : 2

Ans : (b) Let the speed of motorcycle of thief is x Km./hr.

and distance = d km.

Speed of motorcycle of thief $x = \frac{d}{6+4} = \frac{d}{10}$

Speed of motorcycle of Policeman $90 = \frac{d}{4}$

i.e. Distance (d) = 90 × 4 = 360 km.

According to question,

Speed of motorcycle of thief $x = \frac{360}{10}$

x = 36 Km./hr.

25. The radius of a wheel is 0.35 meter. It rotates 450 round in 1 minute. Its speed in (km/hr) is :

(a) 59.40 (b) 59.04
(c) 48.05 (d) 56.40

RRB NTPC 30.04.2016 Shift : 2

Ans : (a) Circumference of wheel $= 2\pi r = 2 \times \frac{22}{7} \times .35$

$= 2.2$ meter

Speed $= \frac{450 \times 2.2}{60} = \frac{450 \times 22}{60 \times 10} \times \frac{18}{5} \text{km/hr} = \frac{594}{10} = 59.4 \text{km/hr}$

26. Krishna cycled a distance of 90 km at a certain speed. If he cycled 3 km slower every hour, he would have taken 5 more hours to reach his destination. What is the speed in Km./hr. at which Kirshna actually cycled?

(a) 7.5 (b) 9
(c) 18 (d) 15

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (b) Let the actual speed = x Km./hr.
According to question,

$$\frac{90}{(x-3)} - \frac{90}{x} = 5$$
$$\Rightarrow \frac{90[x - x + 3]}{x^2 - 3x} = 5$$
$$\Rightarrow \frac{18 \times 3}{x^2 - 3x} = 1$$
$$\Rightarrow 54 = x^2 - 3x$$
$$\Rightarrow x^2 - 3x - 54 = 0$$
$$\Rightarrow x^2 - 9x + 6x - 54 = 0$$
$$\Rightarrow x(x-9) + 6(x-9) = 0$$
$$\Rightarrow (x-9)(x+6) = 0, x = -6 \text{ (invalid) } x = 9 \text{ valid}$$

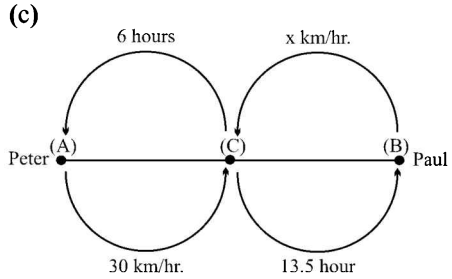
Hence actual speed = 9 Km./hr.

27. Peter belongs to Town A and Paul belongs to Town B. They start their journeys towards each other's towns following the same route at the same time. They meet somewhere on the way and continue with their journeys. After meeting Paul, Peter takes another 13.5 hours to reach his destination while Paul takes another 6 hours to reach Peter's town. If Peter travelled at the speed of 30 Km./hr., what was Paul's speed in Km./hr.?

(a) 42.5 (b) 40
(c) 45 (d) 47.5

RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (c)



Let if,

Speed of Paul is x Km./hr. then,

It is clear from the above diagram that

$$\frac{405}{x} = \frac{6x}{30}$$
$$\Rightarrow 6x^2 = 405 \times 30$$
$$\Rightarrow x^2 = 2025 \text{ Km./hr. or } x = 45 \text{ Km./hr.}$$

28. Geeta travels 120 km by steamer, 450 km by train and 60 km by scooter from Hyderabad to IIT Roorkee. The total journey takes 13 hours and 30 minutes and the speed of the train is 3 times the transit of the scooter and 1½ times that of the same steamer. What is the speed of the train?

(a) 60 Km./hr. (b) 70 Km./hr.
(c) 65 Km./hr. (d) 54 Km./hr.

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (a) According to question,
Let the speed of train = x Km./hr.
speed of train = 3 × speed of scooter
x = 3 × speed of scooter
speed of scooter = x/3 Km./hr.
speed of train = 3/2 × speed of steamer
speed of steamer = 2x/3

$\because$ Speed = $\frac{\text{Distance}}{\text{Time}}$

from question–

$$\frac{120}{\frac{2x}{3}} + \frac{450}{x} + \frac{60}{\frac{x}{3}} = 13 \text{ hour } 30 \text{ minutes}$$
$$\frac{180}{x} + \frac{450}{x} + \frac{180}{x} = \frac{27}{2}$$
$$\frac{810}{x} = \frac{27}{2}$$
$$x = 30 \times 2$$
$$x = 60 \text{ Km./hr.}$$

Hence speed of train = 60 Km./hr.

Type - 2

29. Mark had to cover a distance of 260 km in 4.5 hours using a train traveling at a speed of 70 km/hr or a bus traveling at a speed of 48 km/hr or using both these modes of transport.

(a) 2 hours (b) 1 hour
(c) 1 hour 30 minute (d) 2 hours 30 minute

RRB RPF Constable – 25/01/2019 (Shift-III)

Ans : (d) Let the time taken by bus = t
The time taken by train = (4.5 – t)
According to question,

$$260 = 48 \times t + 70 \times (4.5 - t)$$
$$260 = 48t + (70 \times 4.5 - 70t)$$
$$260 = 48t + 315 - 70t$$
$$22t = 55$$
$$\text{or } \boxed{t = 2.5}$$
$$t = 2 \text{ hours } 30 \text{ minutes}$$

30. Raima covered some distance with a speed of 7 km/h on foot and some distance at a speed of 12 km on a bicycle. He had covered a distance of 64 km in 7 hours. How many hours did he travel on a bicycle?

(a) 2 (b) 3
(c) 5 (d) 4

RRB RPF SI – 11/01/2019 (Shift-I)

Ans. (b) Let Raima covered x_1 distance in t hour by bicycle.
Hence the distance covered by bicycle $(x_1) = 12 \times t$
and distance covered on foot $(x_2) = 7(7 - t)$
so, $x_1 + x_2 = 64$ km.
$12t + 49 - 7t = 64$
$5t = 64 - 49$
$5t = 15$
$t = 3$ hours

31. How long does it take to cross 132meter long bridge by a 110 meter long train running at a speed of 72 km?
(a) 14.3 second (b) 9.8 second
(c) 12.1 second (d) 12.42 second
RRB JE - 23/05/2019 (Shift-II)

Ans : (c)
Length of train (l) = 110 meter
Speed of train = 72 Km./hr. = $72 \times \frac{5}{18} = 20$ m./sec.
Time = T = ?
$\therefore$ Speed of train $= \frac{\text{Distance}}{\text{Time}} = \frac{\ell + \text{length of bridge}}{T}$
$20 = \frac{110+132}{T}$
$T = \frac{242}{20} = \frac{121}{10} = 12.1$ second

32. In a 200 meters race P beats Q by 35 meters or 7 seconds. Find the time taken by P during the race.
(a) 36 second (b) 33 second
(c) 40 second (d) 47 second
RRB JE - 24/05/2019 (Shift-I)

Ans : (b) Time taken to cover 35 m. distance = 7 sec.
$\therefore$ Time taken to cover 200m distance $= \frac{7}{35} \times 200$
= 40 sec.
Hence time taken by Q cover 200 m distance = 40 sec.
Hence time taken by P to covers 200 m distance = 40–7
= 33 sec.

33. At 3/4 of a usual speed, a person reaches his work-place 15 minutes late. Normally how many minutes does it take to reach the work place?
(a) 42 minute (b) 30 minute
(c) 45 minute (d) 60 minute
RRB JE - 24/05/2019 (Shift-III)

Ans : (c) Let the time taken to cover d distance from v speed is t.
$v \xrightarrow{d} t$
$\frac{3}{4} v \xrightarrow{d} (t+15)$
$v \times t = \frac{3}{4} v(t+15)$
$4t = 3t + 45$
$4t - 3t = 45$
$t = 45$ minutes

34. In a 300 meter race, P beats Q by 22.5 meters or 6 seconds. Find the time taken by Q to complete the race.
(a) 86 second (b) 78 second
(c) 76 second (d) 80 second
RRB JE - 27/05/2019 (Shift-II)

Ans : (d) It is clear that Q runs 22.5 meters in 6 sec.
Time taken by Q to cover 300 m distance
$= \left(\frac{6}{22.5} \times 300\right) = \frac{6 \times 3000}{225}$
= 79.99 second = 80 second

35. P and Q run around a circular track of 240 meters at a speed of 15 m/min. and 20 m/min respectively. If they start running together then after how much time they will meet again at the starting point.
(a) 25 minute (b) 36 minute
(c) 30 minute (d) 48 minute
RRB JE - 29/05/2019 (Shift-I)

Ans : (d) The time taken by P to run around a 240 m long circular track. $= \frac{240}{15} = 16$ minute
The time taken by Q to run around a 240 m long circular track $= \frac{240}{20} = 12$ minute
Time of meeting at the starting point again = LCM of 16 and 12 = 48 minute
Hence they will be meet at the starting point again after 48 minutes.

36. A thief rides a bike at a speed of 100 km/h. by running away from the police. The police immediately started chasing at 75 km/h after 1 hour the engine of the police car broke down which took 30 minutes to repair, after this the speed of the car is incresed to 120 km/h, how long the thief was caught?
(a) 2 hours 45 minute (b) 3 hours 40 minute
(c) 5 hours 15 minute (d) 2 hours 30 minute
RRB JE - 02/06/2019 (Shift-I)

Ans : (c) Given-
Speed of thief = 100 Km./hr.
Distance covered by thief in 1 hour = 100 km.
Speed of police = 75 km./hr.
Distance covered by police in 1 hour = 75 km.
An hour later the distance between both = 25 km.
The time it took to recover after the car broke down
= 30 minute
Distance covered by thief in 30 min. = 50 km.
Now distance between both= 25+50 = 75 km.
The time taken by police to catch the thief at the speed of 120 Km./hr. $= \frac{75}{20(\text{Relative speed})}$
$= 3\frac{15}{20} =$ 3 hours. 45 minutes
Hence the total time to catch the thief =
3:45 + 1:30 = 5:15 hours

37. Two friends start walking towards the city P to Q and city Q to P respectively at the same time. After meeting at some point between P and Q, they reach their destination in 54 and 24 minutes respectively. In what time did that friend who went from Q to P finish their journey.

(a) 48 minute (b) 36 minute
(c) 72 minute (d) 60 minute

RRB JE - 26/06/2019 (Shift-III)

Ans : (d) $P\frac{24\,min}{24y}\bullet_{O}\frac{54\,min}{54x}Q$

Where x and y is the speed of two passenger.

$$\sqrt{\frac{t_2}{t_1}}=\frac{x}{y}=\sqrt{\frac{24}{54}}$$

$$\frac{x}{y}=\frac{2}{3}$$

$$x=\frac{2y}{3}$$

$$\text{time}=\frac{24y+54x}{y}=\frac{24y+54\times\frac{2}{3}y}{y}$$

$$=\frac{60y}{y}=60\ \text{min.}$$

38. Rahi had to travel 360 km in 8 hours. During the journey, he traveled at an average speed of 60 km/h but took two breaks in between, the long break doubled from the short time break, for how many minutes the short break was taken.

(a) 30 (b) 45
(c) 40 (d) 35

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (c) Let the short break was taken for x minutes during the considered journey.

Time taken while taking both breaks = x+2x
= 3x minute

Actual speed of Rahi = $\frac{360}{8}=45km/hr$

And Average speed = 60 Km./hr.

According to question,

$$\frac{360}{45}-\frac{360}{60}=\frac{3x}{60}\text{hour}$$

$$\frac{60-45}{45\times60}=\frac{3x}{360\times60}$$

$$15=\frac{3x}{8}$$

3x = 120 minute

x = 40 minute

Hence time taken while taking short break x = 40 minute

39. Nidhi takes 3 hours 45 minutes to walk from one place and return to the same place by bicycle, it takes 4 hours 20 minutes to walk. So how long will it take to get on the cycle.

(a) 3 hours 10 minute
(b) 3 hours 35 minute
(c) 3 hours 45 minute
(d) 3 hours 15 minute

RRB RPF SI – 12/01/2019 (Shift-III)

Ans : (a) Time taken by Nidhi for reaching one side on foot + another side by cycle = 3 hour 45 minutes

$$\Rightarrow 3+\frac{45}{60}=3+\frac{3}{4}=\frac{15}{4}$$

Time taken by her to reach both sides

$$\Rightarrow 4\text{ hours }20\text{ minutes}=4+\frac{20}{60}=\frac{13}{3}$$

Time taken to walk one side $=\frac{13}{3}\times\frac{1}{2}=\frac{13}{6}$

time taken to reach another side by cycle $=\frac{15}{4}-\frac{13}{6}$

$$=\frac{45-26}{12}=\frac{19}{12}\text{hour}$$

time taken to travel both side by cycle $=\frac{19}{12}\times2=\frac{19}{6}$

$$3\text{ hours }\frac{1}{6}\times60=3\text{ hours }10\text{ minutes}$$

40. Hema takes 9 hours 55 minutes to walk a certain distance and return by bicycle. She takes 12 hours and 30 minutes to walk and walk on the same distance, how long does it take to go from both sides by bicycel and return from bicycle.

(a) 7 hour 20 min (b) 7 hour 15 min
(c) 7 hour 35 min (d) 7 hour 45 min

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans :(a) The time taken to travel and reach certain distance on foot and cycle = 9 hour 55 min.

and total time taken in traveling to go and come
= 12 hours 30 minute

∴ time taken in traveling on foot = 6 hour 15 minute

Hence time taken to go and come by cycle

= (9.55–6.15)× 2 = 7 hours 20 minute

41. How long will it take for a student to take a round of 25 hectars field at the speed of 10 km/hr.

(a) 8 min (b) 16 min
(c) 10 min (d) 12 min

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (d) :

∵ 1 Hectare = 10,000 m^2

∴ 25 Hectare = 250,000 m^2
= (500 × 500) m^2

Hence the shape of field is square.

$\therefore$ a = 500 meter

Perimeter of field = 4a = 4 × 500 = 2000 meter

time take to complete one round up = $\frac{\text{Total distance}}{\text{speed}}$

$= \frac{2000\text{ m.}}{10\text{ km./hr.}} = \frac{2000\text{ m.}}{10 \times \frac{1000}{60}\text{ m./minute}}$ =12 minutes

42. I walk a certain distance and return from the vehicle. It takes a total of 13/2 hours., while both it takes 31/4 hours to travel, then how much time will it take to travel both side there.

(a) 6 hours 15 minute
(b) 5 hours 55 minute
(c) 4 hours 15 minute
(d) 5 hours 15 minute

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (d) The time taken by foot (P) and vehicle (w) in go and come = 13/2 hour

Time taken to travel on foot (P) on both sides $= \frac{31}{4}$ hour

Hence

$$P + w = \frac{13}{2}, \qquad P + P = \frac{31}{4}$$

$$2P = \frac{31}{4}, \quad P = \frac{31}{8} \text{ hour}$$

Putting the value of P-

$$\frac{31}{8} + w = \frac{13}{2}$$

$$w = \frac{13}{2} - \frac{31}{8}$$

$$w = \frac{21}{8} \text{ hours}$$

Hence time taken by going vehicle $= \frac{21}{8}$ hours

Time taken in returning by vehicle $= \frac{21}{8}$ hours

Time taken in going and coming both sides by vehicle-

$$\frac{21}{8} + \frac{21}{8} = \frac{42}{8}$$

$\Rightarrow 5\frac{2}{8}$ hours

or 5 hours 15 minutes

43. Kiran has to cover a distance of 300 km in 5 hours. He travels for some time at a speed of 75 km/hr and the remaining the time travels at a speed of 55 km/h. How long did Kiran travel at high speed.

(a) 1 hour 10 minute
(b) 1 hour 25 minute
(c) 1 hour 15 minute
(d) 1 hour 35 minute

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (c) Let t hour time is taken in travelling at the speed of 75 Km./hr.

$75 \times t + 55 \times (5 - t) = 300$

$75t + 275 - 55t = 300$

$20t = 25$

$t = \frac{25}{20} = \frac{5}{4}$

t = 1 hour 15 minute

Hence kiran traveled 1 hours 5 minute at maximum speed.

44. Two cars start at opposite ends of the road and travel towards each other at speeds of 24 km/h and 30 km/h respectively. The distance between the starting positions of the cars is 86.4 km. When the two cars meet each other, the approximate time will be?

(a) 14.4 hours (b) 2.88 hours
(c) 1.6 hour (d) 3.6 hours

RRB Group-D – 10/10/2018 (Shift-I)

Ans : (c) Given–

Total Distance = 86.4 km

Relative speed = 24 Km./hr. + 30 Km./hr.

= 54 Km./hr.

$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{86.4\text{km}}{54\text{km/h}} = 1.6\text{h}$

Time = approximate 1.6 hours

45. Sabrina traveled 8 km/h on foot and 13 km/h on bicycle. She covered 84 km in 8 hours, for how many hours did she ride a cycle.

(a) 4 (b) 3
(c) 5 (d) 2

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (a) Let Sabrina cycled till t time

According to question,

$8(8 - t) + 13t = 84$

$64 - 8t + 13t = 84$

$5t = 20$

$t = 4$

Hence Sabrina cycled till 4 hour.

46. Rahman takes 10 hours to walk to a certain place and return by ride. However, if he had traveled on both sides by ride, he could have saved 5 hours. How long will it take to travel on both sides by walking?

(a) 15 hours (b) 10 hours
(c) 5 hours (d) 20 hours

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (a) Let time taken in walking on foot = tx

Ride has taken time in the travel = ty

According to question,-

$tx + ty = 10$ ____(i)

And, $ty + ty = 5$

$2ty = 5$

$ty = \frac{5}{2}$

from equation (i)

$$tx + \frac{5}{2} = 10$$

$$tx = \frac{15}{2}$$

Time to walk on foot both sides.

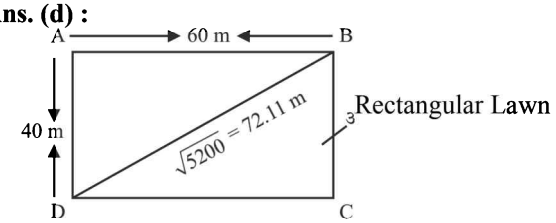
$$tx + tx = 2tx = 2 \times \frac{15}{2} = 15 \text{ hours}$$

47. A rectangular lawn is 60 meters long and 40 meters wide, approximately how long will it take for a person to cross its diagonal at a speed of 3 km/h.

(a) 92.8 second (b) 81.5 second
(c) 84.5 second (d) 86.5 second

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (d) :



Length of rectangular lawn (AB) = 60 m
breadth (AD) = 40 m
Diagonal (DB) = ?

from pythagoras theorem,

$BD^2 = AB^2 + AD^2$
$BD^2 = (60)^2 + (40)^2$
$BD^2 = 3600 + 1600$
$BD^2 = 5200$
$BD = \sqrt{5200}$
BD = 72.11 (Almost72)

Length of diagonal = 72 m
time = ?

$$\text{Speed} = 3\,\text{km/h} = 3 \times \frac{5}{18} = \frac{5}{6}\,\text{m/sec}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{72}{\frac{5}{6}} = \frac{72 \times 6}{5} = 86.4 \text{ (approximately 86.5) sec}$$

48. Ankit leads the church by bicycle and return to the starting point by car. He takes a total of 5 hours for this journey. Using the car on both sides will save him 3 hours in this journey, if he rides on both sides, how much time will it take him?

(a) 5 hours (b) 7 hours
(c) 8 hours (d) 9 hours

RRB Group-D – 26/11/2018 (Shift-III)

Ans. (c) Let the speed of cycle is x and speed of car = y

$$\frac{d}{x} + \frac{d}{y} = 5 \text{.....(i)}$$

$$\frac{2d}{y} = 2 \quad \text{(when traveling both sides by car)}$$

$$\frac{d}{y} = 1 \text{.....(ii)}$$

from equation (i)

$$\frac{d}{x} + 1 = 5$$

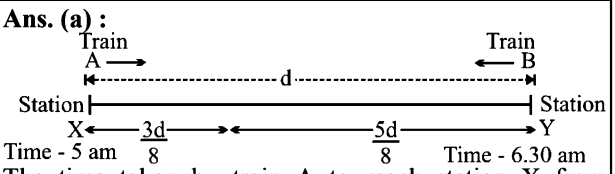
$$\frac{d}{x} = 4 \text{ hour}$$

If he rides a bicycle on both sides, it will take 8 hours.

49. Train A, 5 am. at station X departs and 9 am. arrives at station Y. Another train B 6.30 am. departs from station Y and arrives station X at 10 am. at what time trains meet each other.

(a) 7:40 a.m. (b) 7 a.m.
(c) 8 a.m. (d) 7:30 a.m.

Ans. (a) :



The time taken by train A to reach station X from station Y

$$= 9 - 5 = 4 \text{ hours}$$

The time taken by train B to reach station Y from station X

$$= 10 - 6.30 = \frac{7}{2} \text{ hours}$$

Let the distance between the stations is d km.

Hence speed of train A $= \frac{d}{4}$ Km./hr.

and speed of train B $= \frac{d}{7/2} = \frac{2d}{7}$ Km./hr.

Distance traveled by train A in 3/2 $= \frac{d}{4} \times \frac{3}{2} = \frac{3d}{8}$ km.

Now again the distance between the station $= d - \frac{3d}{8} = \frac{5d}{8}$ km.

Relative speed of both trains $= \frac{d}{4} + \frac{2d}{7} = \frac{7d + 8d}{28}$

$$= \frac{15d}{28} \text{ Km./hr.}$$

$$\text{Relative time} = \frac{\text{Distance}}{\text{Relative Speed}} = \frac{5d \times 28}{8 \times 15d}$$

$$= \frac{28}{24} \text{ hours or 1hours 10 minute}$$

Hence Meeting time for both trains
= 6:30 + 1:10 = 7:40 am

50. Pallav cycling at a speed of 12 km/h and walking at a speed of 3.5 km/h, he takes 8 hours to cover a distance of 45 kilometers, how many hours did he cycle.

(a) 5 (b) 2
(c) 4 (d) 3

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (b) Let Pallav ride the cycle x hour
Distance = Speed × Time
Distance travelled by cycle = 12 × x
Time taken to travel on foot = 8-x
Hence distance travel on foot = $3.5\times(8-x)$
According to question,
$12x+3.5(8-x)=45$
$12x+28.0-3.5x=45$
$8.5x=45-28$
$8.5x=17$
or $85x=170$
$x=\frac{170}{85}=2$ hours

51. Mrs. Vijaya takes 9 hours and 50 minutes to walk from her starting place to a certain destination and return to her starting point by running again. It takes 12 hours and 20 minutes to walk from both the origin and the destination to the destination and again from the destination to the original place, both by running and in how much times she will complete.
(a) 7 hours 45 minute (b) 7 hours 20 minute
(c) 7 hour 15 minute (d) 7 hour 20 minute
RRB NTPC 19.01.2017 Shift : 3

Ans : (d) on foot + by running = 9 hours 50 minutes
on 2 foot = 12 hours 20 miutes
on foot = 6 hours 10 minutes
By running = 9 hour 50 minute – 6 hours 10 minute
By running = 3 hours 40 minute
So, time taken to cover the distance on both sides by running.
= 2 × (3 hours 40 minute)
= 6 hours 80 minute
Total time = 7 hours 20 minutes

52. Keerthana takes 9 hrs 20 minutes in walking a distance and riding back to same place where she started. She could walk both ways in 11 hrs. 15 minutes. The time taken by her to ride back both ways is.
(a) 7 hours 25 minute (b) 7 hours 35 minute
(c) 7 hours 45 minute (d) 7 hours 15 minute
RRB NTPC 11.04.2016 Shift : 2

Ans : (a) Let the time taken to going on foot = x
The time taken to come by riding = y
then x + y = 9.33 (1) (9 hr. 20 min. = 9.33 hr.)
and x + x = 11.25 (2) (time taken to go and come on foot)
2x = 11.25
x = 5.625
from equation (i),
5.625 + y = 9.33
y = 9.33 – 5.625
y = 3.705
Then time take to go and come 2y = 2 × 3.705 = 7.41
Intended time = 7 hours 25 minute

53. A trip from mumbai to pune takes 4 hours 30 minutes to run at a speed of 60 km/h. How much time will it take us to travel at the speed of 15 m/s.
(a) $3\frac{3}{4}$ hours (b) 5 hours
(c) $4\frac{2}{3}$ hours (d) 4 hours
RRB NTPC 18.01.2017 Shift : 2

Ans : (b) Let the distance = x km
Distance covered in 270 minute at a speed of 60 Km./hr.
$\Rightarrow 60\text{ (speed)} = \frac{x\text{ (distance)}}{\frac{270}{60}\text{(time)}} \Rightarrow 60=\frac{60x}{270}$
$\boxed{x=270\text{km}}$
$15\text{m/s}=15\times\frac{18}{5}=54\text{km/h}$
time taken to cover distance of 270 km at a speed of 54 Km./hr.
$\frac{270}{54}=5$ hours

54. Bhanu takes a total of 6 hours 50 minute to walk from one place to another and run back to the starting place. He walks on foot on both sides in 8 hours and 30 minutes. Time taken to run on both sides is?
(a) 5 hours 35 min (b) 5 hours 15 min
(c) 5 hours 10 min (d) 5 hours 45 min
RRB NTPC 17.01.2017 Shift-3

Ans : (c) Let the time taken to cover a distance by running = x
and time taken to cover a distance on foot = y
from question–
$x+y=6\frac{5}{6}h=\frac{41}{6}$.....(i)
$2y=8\frac{1}{2}\Rightarrow y=\frac{17}{4}$
From equation (i)–
$x+\frac{17}{4}=\frac{41}{6}\Rightarrow x=\frac{41}{6}-\frac{17}{4}$
$x=\frac{82-51}{12}=\frac{31}{12}$
∴ time taken to cover a distance both sides by running
$=2\times\frac{31}{12}=\frac{31}{6}=5$ hours 10 minute

55. The distance between two points A and B was covered at a speed of 50 km/h in 5½ hours. How much time can be saved if the speed is increased by 5 km/h.
(a) 5 minute (b) 15 minute
(c) 50 minute (d) 30 minute
RRB NTPC 03.04.2016 Shift : 1

Ans : (d) $S_1 = 50km/h, \quad t_1 = 5\frac{1}{2} = \frac{11}{2}$ hours

$S_2 = (50+5) = 55km/h, \; t_2 = ?$

from formula- $S_1t_1 = S_2t_2$ (distance = speed×time)

$$50 \times \frac{11}{2} = 55 \times t_2$$

$$t_2 = \frac{275}{55} = 5 \text{ hours}$$

saved time $= \left(\frac{11}{2} - 5\right) = \frac{1}{2}$ hours

= 30 minutes

56. Shavya takes 8 hours 20 minutes to walk from one place to the same place. She can both ways in 10 hours 15 minutes. It takes time to come back both ways.

(a) 6 hours 25 minute
(b) 6 hours 35 minute
(c) 6 hours 45 minute
(d) 6 hours 15 minute

RRB NTPC 07.04.2016 Shift : 1

Ans : (a) Time taken for shavya to walk from one place to the same place = 8 hour 20 minutes

$= 8 + \frac{20}{60} = 8 + \frac{1}{3} = \frac{25}{3}$ hours

time taken both ways = 10 hour 15 minutes $= 10 + \frac{15}{60}$

$= \left(10 + \frac{1}{4}\right)$ hour $= \frac{41}{4}$ hour

∴ The time taken to return both ways

$= \left(2 \times \frac{25}{3} - \frac{41}{4}\right)$ hour $= \frac{50}{3} - \frac{41}{4}$

$= \frac{200-123}{12} = \frac{77}{12} = 6\frac{5}{12}$ hour

6 hour $\frac{5}{12} \times 60$ minutes = 6 hour 25 minutes

57. Namrata takes 4 hours 15 minutes to walk from one place to the same place. She can cover both the routes in 5 hours 30 minutes. It takes time to come back from both the paths.

(a) 3 hours
(b) 3 hours 35 minute
(c) 3 hours 45 minute
(d) 3 hours 15 minute

RRB NTPC 07.04.2016 Shift : 3

Ans :(a) Let the time taken to going on foot = x hrs.
time taken to return by vehicle = y hrs.

Hence x + y = 4 hr 15 minutes
x + y = 4.25 hrs .....(1)

Time taken to go and come of foot
x + x = 5.50
2x = 5.50
x = 2.75

putting the value of x in equation (1)
2.75 + y = 4.25
y = 4.25 – 2.75
y = 1.50

Hence time taken to return by vehicle = 1.50 hrs

Then time taken to go and come = 1.50 × 2 = 3 hrs.

58. A bus runs at a speed of 80 km/hr. and reaches its destination with a delay of 10 minutes. If it had moved at a speed of 90 km/hr., it would have reached only 8 minutes late so time is right for completing for journey by bus.

(a) 8 minute (b) 10 minute
(c) 12 minute (d) 15 minute

RRB NTPC 22.04.2016 Shift : 1

Ans : (a) Given–
V_1 = 80 Km./hr. V_2 = 90 Km./hr.
t_1 = 10 minute t_2 = 8 minute

Distance covered by bus from origin to distination

$$= \frac{V_1.V_2}{V_1 \sim V_2} \frac{(t_1 \sim t_2)}{60}$$

$$= \frac{80 \times 90}{80 \sim 90} \times \left(\frac{10-8}{60}\right)$$

$$= \frac{80 \times 90}{10} \times \frac{2}{60} = 24 \text{ Km.}$$

Time taken by bus to complete the journey

$= \frac{24}{80} \times 60 - 10$ = 18 -10 = 8 minute

59. Neelam and manisha start running from the same place at the speed of 7 km/h and 9 km/h respectively, if they move in the same direction how long will they take to go 16 km from each other.

(a) 7 Hours (b) 8 Hours
(c) 10 Hours (d) 12 Hours

RRB NTPC 22.04.2016 Shift : 1

Ans : (b) Speed of Neelam = 7 Km./hr.
Speed of Manisha = 9 Km./hr.

Relative speed of neelam and Manisha = (9–7) Km./hr.
= 2 Km./hr.

Hence, Time taken to go 16 km away from each other

$= \frac{\text{Distance}}{\text{Speed}}$

$= \frac{16}{2}$ = 8 hours

60. Divya takes 4 hours 45 minutes to walk from one place to the same place and she can walk back in 5 hours and 55 minutes. How long will it take for both of them to come back riding.

(a) 3 Hours 35 minute
(b) 3 Hours 55 minute
(c) 3 Hours 45 minute
(d) 3 Hours 15 minute

RRB NTPC 26.04.2016 Shift : 2

Ans : (a) The time taken by Divya to walk from one place and back to the same place by vehicle $= 4 + \frac{45}{60} = \frac{19}{4}$ hours

The time taken to walk from both sides

$$= 5 \text{ hour } 55 \text{ minute} = 5\frac{55}{60} = \frac{71}{12} \text{ hour}$$

The time taken to go on foot from one side

$$= \frac{71}{12} \times \frac{1}{2} = \frac{71}{24}$$

Time taken by both sides from vehicle $= \left(\frac{19}{4} - \frac{71}{24}\right) \times 2$

$$= \left(\frac{114-71}{24}\right) \times 2$$

$$= \frac{43}{24} \times 2 = \frac{43}{12} = 3 \text{ hour } 35 \text{ minute}$$

61. A man at 6:30 am starts walking and wants to travel 30 km. His initial speed is 6 km/h and after traveling 3/5 of distance he reduced his speed to 2 km/h. He will finish his journey:

(a) 11.00 am (b) 12.30 pm
(c) 11.30 pm (d) 12.00 pm

RRB NTPC 27.04.2016 Shift : 2

Ans : (b) Total Distance = 30 km.

Distance traveled at a speed of 6 Km./hr.

$$= \frac{3}{5} \text{ of total distance}$$

$$= 30 \times \frac{3}{5}$$

$$= 6 \times 3$$

$$= 18 \text{ km}$$

On 2 Km./hr. reduction in speed (6 – 2) = 4 Km./hr.

Hence the last 12 km distance will run at a speed of 4 Km./hr.

Hence time taken to cover the entire distnace of 30 km

$$= \frac{18}{6} + \frac{12}{4} = 3 + 3 = 6 \text{ hours}$$

Hence time of ending the journey = 6 : 30 + 6 hours
= 12:30 pm

62. The distance between two places can be covered in $3\frac{1}{2}$ hours at a speed of 62 Km./hr. If the speed is increased by 8 Km./hr., how much time would be saved?

(a) 24 minute (b) 20 minute
(c) 30 minute (d) 15 minute

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (a) Distance = Speed × Time

$$= 62 \times \frac{7}{2}$$

$$= 31 \times 7 = 217 \text{ km.}$$

Time taken to increase speed at 8 Km./hr.

$$\frac{217}{(62+8)} = \frac{217}{70} = \frac{31}{10} = 3\frac{1}{10}$$

Remainig time $= 3\frac{1}{2} - 3\frac{1}{10}$

$$= \frac{7}{2} - \frac{31}{10} = \frac{35-31}{10} = \frac{4}{10} = \frac{2}{5} \text{ hours}$$

$$= \left(\frac{2}{5} \times 60\right) \text{ minute}$$

= 24 minute

63. Raj takes $2\frac{1}{3}$ hours to complete a certain distance at a speed of 51 Km./hr. What time would Kiran take to complete the same distance at a speed of 68 Km./hr.?

(a) $1\frac{2}{3}$ hours (b) $1\frac{3}{4}$ hours
(c) 2 hours (d) $1\frac{1}{2}$ hours

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (b) The total distance covered by Rajan $= 51 \times \frac{7}{3} = 119$ km.

The time take to covered the distance of 119 km by Kiran

$$= \frac{119}{68} \text{ hours}$$

$$= \frac{7}{4} = 1\frac{3}{4} \text{ hours}$$

64. A Car can cover 350 km in 4 hours. If its speed is decreased by $12\frac{1}{2}$ Km./hr. how much time does the car take to cover a distance of 450 km.

(a) 7 hours (b) 4 hours
(c) 5 hours (d) 6 hours

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (d) Initial speed of Car $= \frac{350}{4} = 87.5$ Km./hr.

Speed of Car $12\frac{1}{2}$ = by reducing 12.5 Km./hr.

=87.5–12.5=75 km.hr.

Hence the time taken by Car to covered distance of 450 km

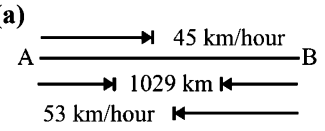
$$= \frac{450}{75} = 6 \text{ hours}$$

65. You are driving at a speed of 45 km/h from point A to B. Your friend is driving 53 km/h from point B to A. If the distance between the two points is 1029 km and you both start at the same time, then how many hours will it take you to cross each other?

(a) 10.5 (b) 9.5
(c) 10 (d) 11

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (a)



Time taken to cross each other

$$= \frac{\text{Distance}}{\text{Relative speed (Sum of speed)}}$$

$$= \frac{1029}{98} = 10.5 \text{ hours}$$

66. Mark had to travel 220 km in 4.5 hours. How long did he travel by bus, with a train running at a speed of 60 km/h or a bus traveling at a speed of 40 km/h or a combination of these two?

(a) 1 hour (b) 2 hour
(c) 1 hour 30 minute (d) 2 hour 30 minute

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (d) : Time taken by bus to travel = t hour
According to question,

$$60(4.5 - t) + 40t = 220$$
$$270 - 60t + 40t = 220$$
$$270 - 20t = 220$$
$$20t = 50$$
$$t = \frac{5}{2} = 2 \text{ hour } 30 \text{ minute}$$

67. Pritish, A train which travels at a speed of 60 km/h or a bus that travels at a speed of 40 km/h or both travels 420 km in 7.5 hr, for how long did he travel by bus.

(a) 1 Hour 30 minute
(b) 2 Hours
(c) 1 Hour
(d) 2 Hours 30 minute

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (a) Let the pritish traveled by bus for t hour
∵ Distance = speed × time
60 × (7.5–t) + 40 × t = 420
450 – 60t + 40 t = 420
30 = 20t

$$t = \frac{30}{20} = \frac{3}{2} \text{ hours}$$

t = 1 hour 30 minute

Type - 3

68. A thief runs by a car from a city located 400 km away at a speed of 60 km/h. Just 30 minutes later police starts chasing the thief at a speed of 80 km/h. How much distance covered by the police to catch the thief?

(a) 70 km. (b) 85 km.
(c) 90 km. (d) 120 km.

RRB JE - 23/05/2019 (Shift-I)

Ans : (d)

Distance covered by thief in 30 minutes $= 60 \times \frac{1}{2}$

= 30 km

Relative speed $= (80 - 60) = 20 \text{km/h}$

Total time to catch the thief $= \frac{30}{20} \text{h} = \frac{3}{2} \text{h}$

Distance covered by the police $= \left(80 \times \frac{3}{2}\right) \text{km}$

= 120 km

69. Two horses travel the same distance at the speed of 10 km/h and 15 km/h respectively. If the first horse takes 12 minutes longer than the second horse then the distance was travelled.

(a) 8 km (b) 6 km
(c) 4 km (d) 2 km

RRB RPF SI – 16/01/2019 (Shift-III)

Ans. (b) : The speeds of both horse are 10 Km./hr. and 15 Km./hr. respectively.

Let distance = x km
Time = Distance / Speed

Time taken by first horse $= \frac{\text{Distance}}{\text{Speed}} = \frac{x}{10}$

Time taken by second horse $= \frac{x}{15}$

According to question,

$$\frac{x}{10} - \frac{x}{15} = \frac{12}{60}$$
$$\frac{3x - 2x}{30} = \frac{12}{60}$$
$$\frac{x}{30} = \frac{12}{60}$$
$$x = 6 \text{ km}$$

70. Apoorva covers equal distance at different speeds of 10 km/h, 20 km/h and 6 km/h and it takes a total of 19 minutes to complete this journey. The total distance travelled by her.

(a) 3 (b) 1
(c) 2 (d) 4

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans : (a) Let the total Distance = 3D km
Time = 19 minutes
∴ According to question,

$$\frac{19}{60} = \frac{D}{10} + \frac{D}{20} + \frac{D}{6} \qquad \left\{ \text{Time} = \frac{\text{Distance}}{\text{Speed}} \right.$$

$$\frac{19}{60} = \frac{6D + 3D + 10D}{60}$$
$$19D = 19$$
$$D = 1\text{km.}$$

Hence Total Distance = 3D = 3 × 1 = 3 km.

71. Azhar can complete a journey in 10 hours. He covers half the journey at a speed of 21 km/h and the rest of the journey at the speed of 24 km/h. What distance (in km) is travelled?

(a) 234 (b) 225
(c) 224 (d) 232

RRB RPF SI – 12/01/2019 (Shift-II)

Ans. (c) Let the total distance covered by Azhar = x km.

From question-

$$\frac{x/2}{21}+\frac{x/2}{24}=10\left[\because \text{Time}=\frac{\text{Distance}}{\text{Speed}}\right]$$

$$\Rightarrow \frac{x}{21}+\frac{x}{24}=20$$

$$\Rightarrow \frac{x}{7}+\frac{x}{8}=20\times 3=60$$

$$\Rightarrow \frac{8x+7x}{56}=60$$

$$\Rightarrow 15x=60\times 56$$

$$x=\frac{60\times 56}{15}=4\times 56, \qquad x=224\,\text{km}.$$

72. Sindhu travels a distance of $20\frac{2}{3}$ km. On foot in $7\frac{3}{4}$ hours. Then how many kilometers did he travel per hour.

(a) $\frac{2}{3}$ km (b) $3\frac{2}{3}$ km
(c) 4km (d) $2\frac{2}{3}$ km

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (d)

Distance covered by Sindhu $=20\frac{2}{3}\text{km}=\frac{62}{3}\text{km}$

and time $=7\frac{3}{4}$ hour $=\frac{31}{4}$ hour

Formula- Speed $=\frac{\text{Distance}}{\text{Time}}$

$$\therefore \text{Speed}=\frac{\frac{62}{3}}{\frac{31}{4}} \Rightarrow =\frac{62}{3}\times\frac{4}{31}=\frac{8}{3}\text{km/hr}$$

$$=2\frac{2}{3}\text{km/hr}$$

73. The usual average speed of a car at a certain distance is 50 km/h on a particular day. The average speed was 1/10 less than the usual average speed, as a result it took 18 minutes longer to finish the journey, how much is this distance of the road in kilometers?

(a) 135 (b) 120
(c) 125 (d) 140

RRB RPF SI – 10/01/2019 (Shift-II)

Ans : (a) First condition–

Let the distance of road $= x$ km

Usual average speed of Car is 50 Km./hr.

Time $=\frac{x}{50}$h

Second condition–

Average speed of a particular day

$$=50-50\times\frac{1}{10}=45\text{ Km./hr.}$$

Time $=\frac{x}{45}$ hours

Now According to question,

$$\frac{x}{45}-\frac{x}{50}=\frac{18}{60}$$

$20x - 18x = 18 \times 15$

$2x = 18 \times 15$

$x = 9 \times 15$

Therefore Distance of road = 135 km

74. Mahima has to travel at a speed of 54 km/h insead of 45 km/h to reach the destination on time due to delay of 12 minutes from her usual time. Find the distance travelled.

(a) 75 km (b) 54 km
(c) 67.5 km (d) 90 km

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (b) : Let the distnace travelled by Mahima during the journey is x.

According to question,

Time = Distance/speed

Here the time is same in both cases-

$$\frac{x}{45}=\frac{x}{54}+\frac{12}{60}$$

or $\frac{x}{45}-\frac{x}{54}=\frac{12}{60}$

or $\frac{54x-45x}{45\times 54}=\frac{12}{60}$

or $\frac{9x}{45\times 54}=\frac{1}{5}$

or $\frac{x}{54}=1$

x = 54 km.

75. A person travels 480 km in 4 hours, some distance travels by airplane and some train. If he travels the entire journey by airplane then he saves 4/5th time than that of the train and he arrives at destination 2 hours before. Find the distance travelled by the train.

(a) 90 km (b) 120 km
(c) 80 km (d) 110 km

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (b) Total time = 4 hours

Total distance = 480 km.

Let the distance covered by train = x km.

and if the total distance covered by the train then the total time taken is = t hours

According to question,

If the total distance is covered by Airplane then the time taken

$$= t - \frac{4t}{5} = \frac{t}{5}$$

$$\frac{t}{5} = (4-2) = 2 \text{ hours}$$

$$\frac{t}{5} = 2$$

t = 10 hours

Hence speed of train $= \frac{480}{10} = 48 km/hr$

and speed of airplane $= \frac{480}{2} = 240 km/hr$

Distance covered by train = x km

So distance covered by airplane = (480–x) km.

Hence

$$\frac{x}{48} + \frac{(480-x)}{240} = 4$$

$$\frac{10x + 960 - 2x}{480} = 4$$

8x = 1920 – 960

$$\boxed{x = 120 \text{ km}}$$

76. A person reaches office driving at a speed of 42 Km/hr 2 minutes earlier. While he drives at a speed of 36 km/h. He reaches 5 minutes late. How much distance does he cover (in km)?

(a) 30.8 (b) 29.6

(c) 30.4 (d) 29.4

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (d) : Let the distance covered by the man is x km.

According to question,

$$\frac{x}{36} - \frac{x}{42} = \frac{7}{60}$$

$$\frac{x}{36} - \frac{x}{42} = \frac{7}{60}$$

$$\frac{42x - 36x}{36 \times 42} = \frac{7}{60}$$

$$\frac{6x}{36 \times 42} = \frac{7}{60}$$

$$x = \frac{7 \times 42 \times 6}{60}$$

$$x = \frac{7 \times 42}{10}$$

x = 29.4 km

77. Sachin completes 360 km distance from Delhi to Hyderabad in 4 hours some by airplane and some by train. If he had travelled the entire journey by airplane he would have saved 4/5 of time than that of the train and reached the destination 2 hours earlier, find the distance travelled by the train.

(a) 45 km (b) 75 km

(c) 50 km (d) 90 km

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (d) Let the distance covered by train is x km and time taken by train is x hour

$$\frac{4x}{5} = 2$$

x = 2.5 hour

It takes less than 2 hours when he travels the entire journey by Airplane.

Hence taken time = 4 – 2 = 2 hours

Speed of Airplane = $\frac{360}{2} = 180$ Km./hr.

According to question,

$$\frac{360 - a}{180} + 2.5 = 4$$

$$\frac{360 - a}{180} = 1.5$$

360 – a = 270

Hence distance (a) = 90 km

78. A woman reaches the office driving at a speed of 40 km/h 4 minutes earlier while she arrives 5 minutes late if she drives at the speed of 32 km/h, what is the distance covered by him?

(a) 32 (b) 24

(c) 28 (d) 30

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (b) Let the actual time to reach the office = t hr

From {Distance = Speed × Time}-

$$40\left(t - \frac{4}{60}\right) = 32\left(t + \frac{5}{60}\right)$$

$$40\left(t - \frac{1}{15}\right) = 32\left(t + \frac{1}{12}\right)$$

$$40t - \frac{40}{15} = 32t + \frac{32}{12}$$

$$40t - 32t = \frac{32}{12} + \frac{40}{15}$$

$$8t = \frac{8}{3} + \frac{8}{3}$$

$$8t = \frac{16}{3}$$

$$t = \frac{2}{3}$$

Total distance = $40\left(\frac{2}{3} - \frac{1}{15}\right) \Rightarrow 40 \times \frac{9}{15} = 24 km$

79. Raghu goes to college at a speed of 4 km/h and returns at a speed of 3 km/h. If he takes 5.95 hrs then what is the distance betweed his home and college?

(a) 10 km (b) 10.5 km
(c) 20 km (d) 10.2 km

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (d) Let the distance between the college and home = x km.
According to question,
Time taken to go + Time taken to come = 5.95 hours

$$\frac{x}{4}+\frac{x}{3}=5.95$$
$$\frac{7x}{12}=5.95$$
$$x=0.85\times12$$
$$\boxed{x=10.2\text{km}}$$

80. A person reaches his office 1 minute before at a speed of 42 km/h, while that person traveling at a speed of 36 km/h reaches the office with a delay of 3 minutes. How much distance does a person travel (in km)?

(a) 12.9 (b) 16.8
(c) 15.4 (d) 18.2

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (b) Let the time taken to reach the office = t hour
According to question,-

$$42\times\left(t-\frac{1}{60}\right)=36\times\left(t+\frac{3}{60}\right)$$
$$\Rightarrow 42t-\frac{42}{60}=36t+\frac{108}{60}$$
$$42t-36t=\frac{108}{60}+\frac{42}{60}$$
$$6t=\frac{15}{6}$$
$$6t=\frac{15}{6}$$
$$t=\frac{5}{12}$$

Distance = speed × time

$$=36\times\left(\frac{5}{12}+\frac{3}{60}\right)$$
$$=36\times\frac{28}{60}=\frac{6\times28}{10}=\frac{168}{10}$$

$\boxed{\text{Distance} = 16.8 \text{ km}}$

81. Chetna went from her home to the city at a speed of 8 km/h, but returned home at the speed of 5 km/h by the same route. It took him 19½ hours to travel on both direction. What is the distance of the city from the house of chetna?

(a) 45 km (b) 60 km
(c) 52.5 km (d) 37.5 km

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (b) Let the distance between Chetna's home and city = x km
According to question,

$$\frac{x}{8}+\frac{x}{5}=\frac{39}{2}$$
$$\frac{5x+8x}{40}=\frac{39}{2}$$
$$\frac{13x}{40}=\frac{39}{2}$$
$$x=\frac{39\times20}{13}$$
$$x = 60 \text{ km}$$

82. If Raj drives 10 km/h more than his usual speed, then he reaches his village 90 minutes before. If raj drives 10 km/h less than the usual speed then reaches his village 150 minutes late. Find the total distance travelled by him.

(a) 600 km (b) 500 km
(c) 400 km (d) 300 km

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (d) Let the usual speed of Raj = x Km./hr.
Both the given conditions are taken for a certain distance.
Hence from $v_1t_1 = v_2t_2$

$$\because \quad (x+10)\times\frac{90}{60}=(x-10)\times\frac{150}{60}$$
$$(x+10)\times\frac{3}{2}=(x-10)\times\frac{5}{2}$$
$$3x+30=5x-50$$
$$2x = 80 \Rightarrow x = 40 \text{ Km./hr.}$$

Again let certain time = t hours then

$$(40+10)\left(t-\frac{90}{60}\right)=(40-10)\left(t+\frac{150}{60}\right)$$
$$50\frac{(2t-3)}{2}=30\frac{(2t+5)}{2}$$
$$100t-150=60t+150$$
$$40t=300\Rightarrow t=\frac{15}{2} \text{ hours}$$

Hence required distance $= 40\times\frac{15}{2}=300$ km.

83. A man travels by train and car to reach his office. If he travels 10 km by car and travels the rest by train then he reaches his office in t hours. If he does the exact opposite of it, he reaches office in (t+0.5) hours. If the speeds of the train and car are 50 km/h and 40 km/h respectively, then how much distance does he cover to reach his office?

(a) 100 km. (b) 80 km.
(c) 120 km. (d) 140 km.

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (c) : Let the Man covered D km. distance to reach the office.

According to question,

$$\frac{10}{40}+\frac{(D-10)}{50}=t \quad .....(i)$$

and

$$\frac{10}{50}+\frac{(D-10)}{40}=t+0.5 \quad .....(ii)$$

from equation (ii) - equation (i),

$$\frac{10}{50}+\frac{D-10}{40}-\frac{10}{40}-\frac{D-10}{50}=t+0.5-t$$

or $$\frac{1}{5}+\frac{D-10}{40}-\frac{1}{4}-\frac{D-10}{50}=0.5$$

$$\frac{40+5D-50-50-4D+40}{200}=0.5$$

$$D-20=100$$

$$\boxed{D=120\text{Km}}$$

84. Two men X and Y travel a distance of 21 km between P and Q at the speed of 3 km/h and 4 km/h respectively. Arriving at Q, Y immediately returns and meet X at R. What is the distance between P and R?

(a) 16 km (b) 17.5 km
(c) 17 km (d) 18 km

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : Let the distance between P and R = x km

∴ Time taken by both will be equal.

time taken by Y = time taken by X

$$\frac{21+21-x}{4}=\frac{x}{3}$$

(42-x)× 3 = 4x

126-3x = 4x

7x= 126

x = 18 ∴ PR = 18 km

85. A bus crosses two persons cycling in the same direction of the bus at a speed of 9.6 km/h and 12 km/h respectively, in 4.5 seconds and 9 seconds respectively. What is the length of the bus?

(a) 6 m (b) 8 m
(c) 7 m (d) 4 m

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (a) Let the length of bus = x meter.

$$\frac{x}{9.6\times\frac{5}{18}}+4.5=\frac{x}{12\times\frac{5}{18}}+9$$

$$\frac{18x}{48}+4.5=\frac{18x}{60}+9$$

$$\frac{18x}{48}-\frac{18x}{60}=9-4.5$$

$$\frac{1080x-864x}{2880}=4.5$$

$$216x=1296$$

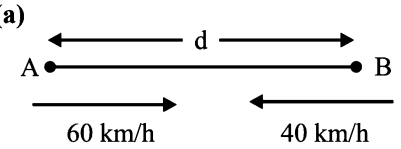
$$x=\frac{1296}{216}=6\text{ m}$$

86. The distance between two points is travelled by the speed of 60 km/h while going and by 40 km/h during the return journey. If it took a total of 5 hours, then the distance between the two points on one side is:

(a) 120 km (b) 135 km
(c) 150 km (d) 180 km

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (a)



$$V_1=\frac{d}{t_1} \qquad V_2=\frac{d}{t_2}$$

$$t_1=\frac{d}{V_1} \text{ and } t_2=\frac{d}{V_2}$$

∵ Total time to come and go = $t_1 + t_2$ = 5h

$$\because 5=\frac{d}{V_1}+\frac{d}{V_2}$$

$$5=\frac{d}{V_1}+\frac{d}{V_2}$$

$$5=\frac{d}{60}+\frac{d}{40}$$

$$5=d\left(\frac{4+6}{240}\right)$$

$$5=d\times\frac{10}{240}$$

d = 5 × 24 = 120 km

∴ d = 120 km

87. Jatin covers 7.5 km more distance if he walks 9 km/h instead of 4 km/h. What is the total distance covered by him (in km)?

(a) 7 (b) 5
(c) 8 (d) 6

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (d) More distance covered after increasing speed from 4 Km./hr. to 9 Km./hr. = 7.5 km

∴ 9 Km./hr. – 4 Km./hr. = 7.5 km

or, 5 Km./hr. = 7.5 km

∴ Time $=\frac{7.5}{5}$hr

∴ Distance $=\frac{7.5}{5}\times 4$ (Distance = speed × time)

= 6 km

Hence total distance of journey = 6 km

88. A car travels a certain distance in 8 hrs. It travels half of the distance at a speed of 40 km/h and another half distance at a speed of 60 km/h. Find the total distance travelled.

(a) 384 kms (b) 368 kms
(c) 344 kms (d) 388 kms

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (a) Let the certain distance covered by the car in 8 hours = d km

$\because$ Covered distance at the speed of 40 Km./hr. $= \frac{d}{2}$ km

$\therefore$ time taken $= \frac{d}{40 \times 2} = \frac{d}{80}$ hr

$\because$ Covered distance at the speed of 60 Km./hr. $= \frac{d}{2}$ km

$\therefore$ time taken $= \frac{d}{60 \times 2} = \frac{d}{120}$ hr.

$\because$ Total time = 8 hr.

$\therefore \frac{d}{80} + \frac{d}{120} = 8$

$\therefore \frac{d}{8} + \frac{d}{12} = 80$

$\frac{3d + 2d}{24} = 80 \Rightarrow 5d = 80 \times 24$

$d = \frac{80 \times 24}{5} = 384$ km

89. The distance between two points is travelled by the speed of 72 km/h while going and by 54 km/h during the return journey. If it took a total of 7 hours, then the distance between the two points on one side is:

(a) 324 km (b) 180 km
(c) 189 km (d) 216 km

RRB Group-D – 22/10/2018 (Shift-II)

Ans. (d) Let the distance between the two points is x km.

According to question, $\frac{x}{72} + \frac{x}{54} = 7$

$\left(\because \text{ Time} = \frac{\text{Distance}}{\text{Speed}}\right)$

$\frac{3x + 4x}{216} = 7$

$7x = 7 \times 216$

x = 216 km

Hence the distance between the two points is 216 km.

90. Travelling at a speed of 57 km/h Manav reached the destination 3 minute earlier. If he travelled at the speed of 51 km/h then he reached 1 minute late. What is the distance travelled by Manav?

(a) 31.5 km (b) 32.3 km
(c) 31.9 km (d) 32.8 km

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (b)

Let the total distance is D km. and usual time is t.

Hence $\frac{D}{57} = t - \frac{3}{60}$.....(1)

and $\frac{D}{51} = t + \frac{1}{60}$.....(2)

By subtracting equation (1) from equation (2),

$\frac{D}{51} - \frac{D}{57} = \frac{1}{60} + \frac{3}{60}$

$\frac{6D}{51 \times 57} = \frac{4}{60} = \frac{1}{15}$

$D = \frac{51 \times 57}{6 \times 15} = 32.3$km

91. A woman reaches the office 5 minutes earlier driving at a speed of 45 km/h, while she reaches 7 minutes late woman driving at a speed of 27 km/h. What distance did she travel?

(a) 18 (b) 16.5
(c) 15 (d) 13.5

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (d) : The distance travelled by the woman in both conditions will be the same {$\because$ t = time taken to arrive at the right time}

$v_1t_1 = v_2t_2$

$45 \times \left(t - \frac{5}{60}\right) = 27 \times \left(t + \frac{7}{60}\right)$

$\frac{5(60t - 5)}{60} = \frac{3(60t + 7)}{60}$

$5(60t - 5) = (60t + 7) \times 3$

300t–25 = 180t + 21

120 t = 46

$t = \frac{46}{120} = \frac{23}{60}$ hours.

Putting the value of t-

$\therefore$ Distance $= 45 \times \left(\frac{23}{60} - \frac{5}{60}\right)$

$= 45 \times \frac{18}{60}$

$= \frac{135}{10} = 13.5$km

92. Ram and Lakshman travel the same distance at a speed of 32 km/h and 40 km/h respectively. Ram takes one hour more than Lakshman to cover the distance. How much distance travelled by lakshman?

(a) 160 km (b) 200 km
(c) 120 km (d) 100 km

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (a) Let Distance = D km.

Speed of Ram = 21 Km./hr.

Speed of Laxman = 40 Km./hr.

According to question,

$\frac{D}{32} - \frac{D}{40} = 1$

$\frac{40D - 32D}{1280} = 1$

8D = 1280

D = 160 km.

93. Travelling at a speed of 66 Km/hr, Piyali reaches a place 5 minutes earlier. If she travels at the speed of 54 km/h, she is delayed by 3 minutes. How much distance did Piyali travel?

(a) 39.2 km (b) 40.2 km
(c) 39.6 km (d) 40.8 km

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (c) Let the time taken by Piyali to cover the journey is t hours.

∴ Distance from first condition

$$= 66\times\left(t-\frac{5}{60}\right) \quad ....(i)$$

Distance from second condition

$$= 54\times\left(t+\frac{3}{60}\right) \quad ....(ii)$$

By solving equation (i) and (ii)–

$$66\left(t-\frac{5}{60}\right)=54\left(t+\frac{3}{60}\right)$$

$$11\left(t-\frac{5}{60}\right)=9\left(t+\frac{3}{60}\right)$$

$$11t-\frac{55}{60}=9t+\frac{27}{60}$$

$$11t-9t=\frac{27}{60}+\frac{55}{60}$$

$$2t=\frac{82}{60}$$

$$t=\frac{41}{60}$$

Putting the value of t in equation (i),

Distance covered by Piyali $=66\left(\frac{41}{60}-\frac{5}{60}\right)$

$$=66\times\frac{36}{60}=\frac{11\times18}{5}=39.6\,\text{km}$$

94. Rahim started his journey with a delay of 15 minutes due to which he had to drive a car at a speed of 54 km/h instead of 45 km/h to reach his destination on time. Find the distance travelled during this journey.

(a) 75.5 km (b) 90 km
(c) 67.5 km (d) 13.5 km

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (c) Let the covered distance during the journey = d km

According to question,

$$\text{Speed}=\frac{\text{Distance}}{\text{Time}}$$

$$\frac{d}{45}-\frac{d}{54}=\frac{15}{60}$$

$$\Rightarrow \frac{6d-5d}{270}=\frac{1}{4}$$

$$d=\frac{270}{4}=67.5\text{km}$$

95. Vishnu covers the same distance at a speed of 10 km/h, 30 km/h and 8 km/h and takes a total of 15.5 minutes to complete journey, then find the total distance travelled by him.

(a) 1 (b) 3
(c) 4 (d) 2

RRB NTPC 17.01.2017 Shift-1

Ans : (b) Let Vishnu has travelled a total of x km distance of journey.

According to question,

$$\frac{\left(\frac{x}{3}\right)}{10}+\frac{\left(\frac{x}{3}\right)}{30}+\frac{\left(\frac{x}{3}\right)}{8}=\frac{15.5}{60}$$

$$\frac{x}{3}\left(\frac{1}{10}+\frac{1}{30}+\frac{1}{8}\right)=\frac{15.5}{60}$$

$$\frac{x}{3}\left(\frac{24+8+30}{240}\right)=\frac{15.5}{60}$$

$$\frac{x}{3}\left(\frac{62}{240}\right)=\frac{15.5}{60}$$

$$x=\frac{15.5\times240\times3}{60\times62}$$

$$x=\frac{186}{62}$$

x = 3 km.

96. P travels 70 hours. He travels half of his journey at 30 km/h and the remaining half travels at a speed of 40 km/h. Find the total distance of his journey.

(a) 2000 km. (b) 2400 km.
(c) 2720 km. (d) 2160 km.

RRB NTPC 04.04.2016 Shift : 3

Ans. (b) Let the total distance of travelled journey = x km

From question,

$$\frac{x/2}{30}+\frac{x/2}{40}=70$$

$$\Rightarrow \frac{x}{30}+\frac{x}{40}=70\times2=140$$

$$\Rightarrow \frac{4x+3x}{120}=140$$

$$\Rightarrow 7x=140\times120$$

$$\Rightarrow x=\frac{140\times120}{7}=2400\,\text{km.}$$

97. M and N cover the same distance at 160 km/h and 200 km/h respectively, if M takes 30 minutes more than N then the distance is travelled by each.

(a) 120 km. (b) 200 km.
(c) 320 km. (d) 400 km.

RRB NTPC 04.04.2016 Shift : 3

Ans : (d) Let the distance covered by M and N = x km.

$\therefore$ According to question,

$$\frac{x}{160}-\frac{x}{200}=\frac{30}{60}=\frac{1}{2}$$

$$\Rightarrow \frac{x}{20\times 8}-\frac{x}{20\times 10}=\frac{1}{2}$$

$$\Rightarrow \frac{x}{8}-\frac{x}{10}=\frac{1}{2}\times 20=10$$

$$\Rightarrow \frac{5x-4x}{40}=10$$

$$\Rightarrow x=10\times 40=400 \text{ km.}$$

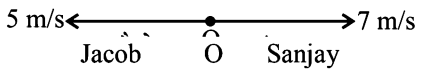
98. Sanjay and Jaicob start running in the opposite direction from the same point of 7 m/s and 5 m/s respectively. How far will they be from each other after 42 minutes?

(a) 30.24 km. (b) 504 km.
(c) 8.4 km. (d) 69.5 km.

RRB NTPC 03.04.2016 Shift : 1

Ans : (a) Distance covered by Jacob in 47 minutes = 5 m/sec × 42 minutes

= 5×42 × 60 m. = 12600 m.



Distance covered by Sanjay in 42 minutes

= 7 m./sec.×42 minute = 7 × 42 × 60 m.

= 17640 m.

Hence required distance between Sanjay and Jaicob after 42 minutes

= 17640 + 12600

= 30240 m. = 30.24 km.

99. A person walks at a speed of 3 km/h. He doubles his speed after reaching the halfway point. He walks for 5 hours. Find the total distance travelled by him.

(a) 18 km. (b) 15 km.
(c) 20 km. (d) 9 km.

RRB NTPC 30.03.2016 Shift : 1

Ans : (c) Let the total distance = x km.

$\because$ Time $=\frac{\text{Distance}}{\text{Speed}}$

$$\frac{\frac{x}{2}}{3}+\frac{\frac{x}{2}}{6}=5$$

$$\frac{x}{6}+\frac{x}{12}=5$$

$$\frac{2x+x}{12}=5$$

$$3x=60$$

$$x=20$$

Total distance = 20 km.

100. If Lalita leaves for her school from her home at a speed of 45 km/h by car then she takes 5 minutes more than driving at 60 km/h. What is the distance between home and school?

(a) 18km (b) 15km
(c) 14km (d) 10km

RRB NTPC 18.01.2017 Shift : 2

Ans : (b) Let the distance of school from home = x km.

then, the time taken to cover the distance of x km at the speed of 45 Km./hr. $=\frac{x}{45}$

time taken to cover the distance of x km at the speed of 60 Km./hr. $=\frac{x}{60}$

According to question,

$$\Rightarrow \frac{x}{45}-\frac{x}{60}=\frac{5}{60}$$

$$\Rightarrow \frac{4x-3x}{180}=\frac{1}{12}$$

$$\Rightarrow x=\frac{180}{12}=15$$

$$\boxed{x=15\text{km}}$$

Distance between School and home = 15 km

101. Ram completes the 4/9th part of the total journey by Bus, the 5/18th part by train and the remaining 10 km by walking. Find the total distance.

(a) 42 km. (b) 90 km.
(c) 36 km. (d) 18 km.

RRB NTPC 19.04.2016 Shift : 2

Ans : (c) Let the total distance is x km.

Distance travelled by bus $=\frac{4x}{9}$

Distance travelled by train $=\frac{5x}{18}$

According to question,

$$x-\left(\frac{4x}{9}+\frac{5x}{18}\right)=10$$

$$\Rightarrow x-\frac{13x}{18}=10$$

$$\Rightarrow \frac{5x}{18}=10$$

$$\Rightarrow 5x=10\times 18$$

$$\Rightarrow x=\frac{10\times 18}{5}=36 \text{ km.}$$

102. A runner covers a distance of 60 km in 3 hours and 45 minutes. How much should he increase the average speed to cover this distance 45 minutes before?

(a) 16 Km./hr. (b) 20 Km./hr.
(c) 6 Km./hr. (d) 4 Km./hr.

RRB NTPC 18.04.2016 Shift : 3

Ans : (d) Given–

Distance = 60 km, Time = 3 hours 45 minutes

$= 3\frac{45}{60}$ hours

$= \frac{15}{4}$ hours

$\therefore$ Speed $= \frac{\text{Distance}}{\text{Time}}$

$= \frac{60}{(15/4)} = 60 \times \frac{4}{15} = 16$ Km./hr.

Let the averge speed be increased by x Km./hr.

time = 3 hours

Speed = (16+x) Km./hr.

According to question,

$(16 + x) = \frac{60}{3}$

$\Rightarrow$ 16+x = 20

$\Rightarrow$ x = 20–16

$\Rightarrow$ x = 4 Km./hr.

103. Mohan covers a certain distance from his car in 12 hours at a uniform speed. When the speed is increased by 5 km/h, then the same distance can be covered in 9 hours. What is the total distance?

(a) 108 km. (b) 90 km.
(c) 190 km. (d) 180 km.

RRB NTPC 18.04.2016 Shift : 3

Ans : (d) Let the total distance is d and the speed is 'V'-

According to first condition-

$V = \frac{d}{12}$.....(i)

According to second condition-

$(V + 5) = \frac{d}{9}$.....(ii)

from equation (i) and (ii)–

$\frac{d}{12} + 5 = \frac{d}{9}$

$\frac{d + 60}{12} = \frac{d}{9}$

$\Rightarrow$ 3d + 180 = 4d

$\Rightarrow$ d =180 km.

104. A bus is running at a speed of 90 km/h. How much distance will it cover in 20 seconds?

(a) 500 meter (b) 450 meter
(c) 180 meter (d) 600 meter

RRB NTPC 16.04.2016 Shift : 1

Ans : (a) Speed of bus = 90 Km./hr.

$= 90 \times \frac{5}{18}$

= 25 m./sec.

$\therefore$ Distance covered in 20 seconds = 20×25 = 500 m.

105. Chandan covers equal distances at the speed of 3 km/h, 4 km/h and 8 km/h respectively and takes a total time of 42.5 minutes. Find the total distance in km.

(a) 4 (b) 2
(c) 1 (d) 3

RRB NTPC 11.04.2016 Shift : 2

Ans : (d) Time (t) = 42.5 minute $= \frac{42.5}{60}$ hours.

Let distance = d, $d = d_1 + d_2 + d_3$

and $d_1 = d_2 = d_3 = x$

Average speed = V

V_1 = 3 Km./hr. V_2 = 4 km/ hr., V_3 = 8 Km./hr.

Average speed = (V) $= \frac{d_1 + d_2 + d_3}{\frac{d_1}{v_1} + \frac{d_2}{v_2} + \frac{d_3}{v_3}}$

$V = \frac{x + x + x}{\frac{x}{3} + \frac{x}{4} + \frac{x}{8}}$

$V = \frac{3x}{\frac{8x + 6x + 3x}{24}}$

$V = \frac{3x}{\frac{17x}{24}}$

$V = \frac{3 \times 24}{17}$ km/hr.

Then,

Distance = speed × time

$d = \frac{42.5}{60} \times \frac{3 \times 24}{17}$

$d = \frac{425}{600} \times \frac{3 \times 24}{17}$

d = 3 km

106. P travels 35 hours. He travels half of the journey at 30 km/h and the rest at 45 km/h. Find the total distance.

(a) 1000 km. (b) 1200 km.
(c) 1360 km. (d) 1080 km.

RRB NTPC 10.04.2016 Shift : 3

Ans : (b) Let the total distance = d km.

Time $= \frac{\text{Distance}}{\text{Speed}}$

$\therefore$ According to question,

$\frac{\frac{d}{2}}{30} + \frac{\frac{d}{2}}{40} = 35$

$\Rightarrow \frac{d}{60} + \frac{d}{80} = 35$

$\Rightarrow \frac{4d + 3d}{240} = 35$

$\Rightarrow$ 7d = 240 × 35

d = 240 × 5 = 1200

107. Rajneesh covers the same distance with the speed of 6 km/h, 4 km/h and 8 km/h respectively and takes a total time of 32.5 minutes. Find the total distance in km.

(a) 4 (b) 2
(c) 1 (d) 3

RRB NTPC 07.04.2016 Shift : 3

Ans : (d) Rajneesh travels the same distance each time.
Let the distance travelled each time = x km

Then average speed $(v) = \dfrac{d_1+d_2+d_3}{\dfrac{d_1}{v_1}+\dfrac{d_2}{v_2}+\dfrac{d_3}{v_3}}$

$$= \frac{x+x+x}{\frac{x}{6}+\frac{x}{4}+\frac{x}{8}} = \frac{3x}{\frac{4x+6x+3x}{24}}$$

$$= \frac{72x}{13x} = \frac{72}{13} \text{ km/hr.}$$

Time = 32.5 minute $= \dfrac{32.5}{60}$ hours

Distance = speed × time $= \dfrac{72}{13} \times \dfrac{32.5}{60} = 3\text{km}$

108. Two buses X and Y travel from the same place in the same direction at a speed 20 km/h and 50 km/h respectively. But Y leaves 6 hours after the X leaves. At what distance will both buses meet from the starting point?

(a) 100 km. (b) 200 km.
(c) 150 km. (d) 300 km.

RRB NTPC 19.01.2017 Shift : 2

Ans : (b) Let y will catch x after t hours.
Distance covered by x in 6 hours = 120 km
According to question,
Relative speed = (50 – 20) = 30 Km./hr.

time taken to catch $= \dfrac{\text{Total distance}}{\text{Relative speed}} = \dfrac{120}{30} = 4$ hrs.

Then–
⇒ Distance covered by y in 4 hours = 4 × 50 = 200 km
⇒ Distance covered by x in 4 hours = 4 × 20 = 80 km
⇒ Total distance covered by x = 80 + 120 = 200 km
⇒ Then both buses will meet at a distance of 200 km. their starting point.

109. Monal Kumar travels the same distance at the speed of 3 km/h, 5 km/h and 8 km/h respectively and takes a total time of 395 minutes. Find the total distance in km.

(a) 40 (b) 20
(c) 10 (d) 30

RRB NTPC 22.04.2016 Shift : 2

Ans : (d) Let the total distance is 3d km.
According to question,

$$\frac{d}{3}+\frac{d}{5}+\frac{d}{8}=\frac{395}{60}$$

$$\frac{40d+24d+15d}{120}=\frac{395}{60}$$

$$\frac{79d}{120}=\frac{395}{60}$$

$$d = 10$$

Hence total distance = 3d = 3 × 10 = 30 km.

110. A car covers the journey from chennai to Coimbatore in 8 hours. The first half of distance of the journey is travelled by a speed of 73 km/h and the second half is travelled at a speed of 81 km/h. What is the distance from Chennai to Coimbatore?

(a) 614.34 km. (b) 600 km.
(c) 592 km. (d) 632.65 km.

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (a) Let the distance is d km.
According to question,

$$\frac{(d/2)}{73}+\frac{(d/2)}{81}=8$$

$$\Rightarrow \frac{d}{146}+\frac{d}{162}=8$$

$$\Rightarrow 308d = 8 \times 146 \times 162$$

$$\Rightarrow d = \frac{8\times146\times162}{308}$$

$$\Rightarrow d = 614.337$$

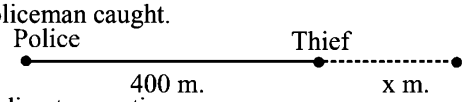
$$\Rightarrow d = 614.34 \text{ km.}$$

111. A thief is 400 meters ahead of a policeman. The thief runs away and policeman starts chasing together. Suppose the speed of the thief is 10 km/h and the speed of the police is 15 km/h. Calculate the distance the thief covered before the policeman caught.

(a) 750 meter (b) 800 meter
(c) 850 meter (d) 900 meter

RRB NTPC 29.04.2016 Shift : 2

Ans : (b) Let the thief covered a distance of x m before the policeman caught.



According to question,

$$\frac{x+400}{15\times\frac{5}{18}} = \frac{x}{10\times\frac{5}{18}}$$

$$\Rightarrow \frac{6x+2400}{25} = \frac{9x}{25}$$

$$\Rightarrow 3x = 2400$$

$$\Rightarrow x = 800\text{m}$$

112. Jai travels from his home to school at a speed of 10 km./hr and reaches late by 5 minutes. If he increases his speed by 3 km./hr, he reaches the school 4 minutes early. What is the distance between his home and the school?

(a) 2 km. (b) 6.5 km.
(c) 4.8 km. (d) 2.5 km.

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (b) Let the covered distance = d

$\because$ d= vt

According to question,

$$d = 10\times\left(t+\frac{5}{60}\right) \quad \text{..... (i)}$$

and $$d = 13\times\left(t-\frac{4}{60}\right) \quad \text{.....(ii)}$$

from equation (i) and (ii),

$$10\times\left(t+\frac{5}{60}\right) = 13\times\left(t-\frac{4}{60}\right)$$

$$10t+\frac{50}{60} = 13t-\frac{52}{60}$$

$$\frac{50}{60}+\frac{52}{60} = 13t-10t$$

$$\frac{102}{60} = 3t$$

$$t = \frac{34}{60}$$

putting the value of t in equation (i)

$$d = 10\times\left(\frac{34}{60}+\frac{5}{60}\right)$$

$$d = 10\times\frac{39}{60} \Rightarrow d = 6.5 \text{ km.}$$

113. A mini van is going in the same direction in which direction two people are going at the speed of 4.8 km/h and 6 km/h. Mini van crosses both people in 4.5 seconds and 9 seconds respectively. What is the length of the mini van?

(a) 3.5m (b) 3 m.
(c) 4 m. (d) 2 m.

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (b) Let the length of the Van is 'a' meter and speed of Van is x Km./hr.

According to question,

$$\frac{a}{(x-4.8)\times\frac{5}{18}} = 4.5$$

$$\frac{18a}{5} = 4.5x - 21.6 \quad \text{.....(i)}$$

$$\frac{a}{(x-6)\frac{5}{18}} = 9$$

$$\frac{18a}{5} = 9x - 54 \quad \text{.....(ii)}$$

from equation (i) and (ii),

x = 7.2 m./sec.

from equation (ii),

$$\frac{18a}{5} = 9\times 7.2 - 54$$

$$\Rightarrow \frac{18a}{5} = 10.8 \Rightarrow a = \frac{10.8\times 5}{18} = 3 \text{ meter}$$

114. A 150 m long train, travelling at the speed of 54 km./hr, crosses a platform in 42 seconds. What is the length of the platform?

(a) 540 m (b) 630 m
(c) 780 m (d) 480 m

RRB ALP & Tec. (10-08-18 Shift-I)

Ans : (d) Let the length of Platform = x meter

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$54\times\frac{5}{18} = \frac{150+x}{42}$$

$\Rightarrow$ x+150 = 3×5×42

$\Rightarrow$ x+150 = 630

$\Rightarrow$ x= 630–150

x = 480

Hence length of platform = 480 m

Type - 4

115. A man walks from P to Q at a speed of 5 km/h and comes back at a speed of 3 km/h from Q to P. What is his average speed during the entire journey?

(a) 8 Km./hr. (b) 3.25 Km./hr.
(c) 3.75 Km./hr. (d) 3.5 Km./hr.

RRB JE - 22/05/2019 (Shift-III)

Ans : (c)

$$\text{Average speed} = \frac{2\times\text{product of speed}}{\text{sum of speed}}$$

$$= \frac{2\times 5\times 3}{5+3} = \frac{30}{8} = 3.75 \text{ Km./hr.}$$

116. A person travels at a speed of 50 km/h from P to Q in 30 minutes. Then he increases his speed to 60 km/h and reaches his destination in 20 minutes. Find the average speed during the entire journey.

(a) 55km. /Hour (b) 54Km./hr.
(c) 65Km./hr. (d) 70Km./hr.

RRB JE - 26/05/2019 (Shift-III)

Ans : (b) Average Speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{50\times\frac{30}{60}+60\times\frac{20}{60}}{\frac{30+20}{60}} \quad (\because \text{distance} = \text{speed}\times\text{time})$$

$$= \frac{25+20}{\frac{5}{6}} = \frac{45\times 6}{5} = 54\text{km/hr.}$$

117. A cart covers the first 10 km. with a speed of 4 km/h and the other 10 km covers at a speed of 2 km/h. Find the average speed of that car in km/h.

(a) 2.67 Km./hr. (b) 3.33 Km./hr.
(c) 2 Km./hr. (d) 5.54 Km./hr.

RRB RPF SI – 16/01/2019 (Shift-III)

Ans : (a) When someone travels a certain distance at the speed of x Km./hr.
and travels the same distance at the speed of y Km./hr., then the average speed in the entire journey

$$= \frac{2xy}{(x+y)} km/h = \frac{2\times4\times2}{4+2} km/h$$

$$= \frac{8}{3} = 2.67 \ km/h$$

118. An object travels 24 m in 3s and 15 m in 2s. What is the average speed of the object?
(a) $6.67ms^{-1}$ (b) $7.8ms^{-1}$
(c) $7.8S^{-1}$ (d) 8.0m

RRB RPF SI – 11/01/2019 (Shift-I)

Ans. (b) Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{(24+15)}{(3+2)} = \frac{39}{5} = 7.8 m/sec = 7.8 \ ms^{-1}$$

119. An object travels 23m in 3s and 15m in 2s. What is the average speed of the object?
(a) $7.6 \ ms^{-1}$ (b) $7.6 \ s^{-1}$
(c) 7.6 m (d) $8.0 \ ms^{-1}$

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans. (a) Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{23+15}{3+2} = \frac{38}{5} = 7.6 ms^{-1}$$

120. Kiran swims in a 90 m long bridge. From one end to the other end then swims back to the same straight path and she covers total distance of 360 m. Find the average velocity of Kiran.
(a) $0 \ ms^{-1}$ (b) $3 \ ms^{-1}$
(c) $5 \ ms^{-1}$ (d) $4 \ ms^{-1}$

RRB RPF SI – 10/01/2019 (Shift-III)

Ans : (a) Since Kiran is coming back to the initial point. Therefore her displacement will be zero. Hence average velocity will be = $0 \ ms^{-1}$

121. An object travels 10 m in 4s and then 14m in 2s. What is the average speed of the object?
(a) $4 \ s^{-1}$ (b) 4 m
(c) $4.5 \ ms^{-1}$ (d) $4 \ ms^{-1}$

RRB RPF Constable – 25/01/2019 (Shift-III)

Ans : (d)
Total distance travelled by the object
= 10m+14m = 24m
and total time taken to cover that distance
= 4sec + 2 sec = 6 sec
Average speed of object = ?

Average speed $= \frac{\text{Total Covered Distance}}{\text{Total Time taken}} = \frac{d_1+d_2}{t_1+t_2}$

$$= \frac{(10+14)m}{(4+2)sec} = \frac{24}{6} = 4ms^{-1}$$

Hence the average speed of object is $4 \ ms^{-1}$

122. A car travels at a speed of 62 km/h for $2\frac{1}{2}$ hours and 68 km/h for $1\frac{1}{4}$ hours. What will be its average speed in total distance travelled?
(a) 65 (b) 64
(c) 63 (d) 61

RRB NTPC 28.04.2016 Shift : 1

Ans : (b) Total distance covered by Car

$$= 62\times\frac{5}{2} + 68\times\frac{5}{4} \quad (\because \text{Distance} = \text{Speed}\times\text{Time})$$

= 31 × 5 + 17 × 5
= 155 + 85 = 240 km.

Average speed of car $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{240}{5/2+5/4} \Rightarrow \frac{240\times4}{10+5}$$

$$= \frac{240\times4}{15} \Rightarrow 16\times4 = 64 \text{ Km./hr.}$$

123. Ramlal runs 18 km at a speed of 9 km/h. At what speed does he need to run in the next 3 hours to get an average speed of 6 km/h in the entire race?
(a) 6 Km./hr. (b) 12 Km./hr.
(c) 4 Km./hr. (d) 9 Km./hr.

RRB NTPC 22.04.2016 Shift : 3

Ans : (c) Time taken to run 18 km $= \frac{18}{9} = 2$ hours

According to question,

$$6 = \frac{18+x}{2+3}$$

x = 30–18 = 12 km.

$\therefore$ Speed $= \frac{12}{3} = 4$ Km./hr.

124. Rupa goes to her office at half of speed at which she returns from her office. The average speed during the entire journey is 24 km/h. What was her speed when Rupa was going to her office.
(a) 18 Km./hr. (b) 72 Km./hr.
(c) 9 Km./hr. (d) 24 Km./hr.

RRB NTPC 22.04.2016 Shift : 3

Ans : (a) Let the Actual Speed of Rupa = S Km./hr.

Average speed $= \frac{2V_1V_2}{V_1+V_2}$

$$\therefore 24 = \frac{2\times S\times\frac{S}{2}}{S+\frac{S}{2}}$$

$$\Rightarrow 12 = \frac{S^2}{3S}$$

$\therefore$ S = 36 Km./hr.
Hence speed at the time on coming to office

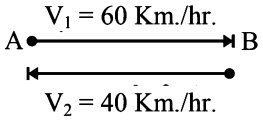
$$= \frac{36}{2} = 18km/h$$

125. The speed of a car from A to B is 60 km/h and the speed of return is 40 km/h. Find the average speed of the car (in km/h).

(a) 50 (b) 45
(c) 48 (d) 52

RRB NTPC 12.04.2016 Shift : 1

Ans : (c)



Average speed of Car $= \frac{2V_1V_2}{V_1+V_2}$

$= \frac{2\times60\times40}{60+40} = \frac{4800}{100} = 48$ Km./hr.

126. An object moves 15 m in 4s and next 16m in 2s. What is the average speed of that object?

(a) 6.17 ms^{-1} (b) 5.17 m
(c) 5.17 s^{-1} (d) 5.17 ms^{-1}

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (d) Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$= \frac{15+16}{4+2} = \frac{31}{6} = 5.17\ ms^{-1}$

127. An object travels 18 m in first 4s and the another 16 m in 2s. What is the average speed of the object?

(a) 5.67 m (b) 5.67 s^{-1}
(c) 5.67 ms^{-1} (d) 6.67 ms^{-1}

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (c) : Distance covered in first 4 seconds = 18 m.
Distance covered in next 2 seconds = 16 m.
Total covered distance = 18+16 = 34 m.
Total time taken = 4+2= 6 seconds

Average speed of object = $= \frac{\text{Total Covered Distance}}{\text{Total Time Taken}}$

Average speed = $\frac{34}{6}$ = 5.67 m./sec.

128. A man covers the distance from point A to B at a speed of 40 km/h and returns from B to A at a speed of 60 km/h. What is his average speed during the entire journey?

(a) 48 Km./hr. (b) 55 Km./hr.
(c) 50 Km./hr. (d) 45 Km./hr.

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (a) Going speed from point A to B = 40 Km./hr.
Returning Speed from point B to A = 60 Km./hr.

From average speed $= \frac{2xy}{x+y}$,

$= \frac{2\times40\times60}{40+60}$

Average speed = 48 Km./hr.

129. An object travels first 25m in 4s and then next 15m in 2s. What is the average speed of the object?

(a) 6.67 m (b) 6.67 s^{-1}
(c) 6.67 ms^{-1} (d) 6.67 ms^{1}

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (c) Distance travelled in 4 seconds = 25 m.
Distance travelled in 2 seconds = 15 m.

Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$= \frac{25+15}{4+2} = \frac{40}{6} = 6.67\ ms^{-1}$

130. Ram swims in a 72 m long pond. He covers a distance of 144 m in 1 minute by swimming from one end to the other and coming back the same way. Find his average speed.

(a) 3.67 ms^{-1} (b) 2.60 ms^{-1}
(c) 2.04 ms^{-1} (d) 2.4 ms^{-1}

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d) Average speed $= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{144}{60}$

$= \frac{12}{5} = 2.4\,ms^{-1}$

131. Ram swims in a 95 m long pond. He travels 190 m in 1 minute by swimming from one end to the other and coming back the same way. Find his average speed.

(a) 2.05 ms^{-1} (b) 3.17 ms^{-1}
(c) 2.00 ms^{-1} (d) 3.10 ms^{-1}

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (b) Average speed $= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{190\text{m}}{1\text{min}}$

$\Rightarrow \frac{190\text{m}}{60\text{Sec}} = \frac{19}{6} = 3.1666$
$= 3.17\ ms^{-1}$

132. Ram swims in 80 m long pond. He travels 160 m in 1 minute by swimming from one end to the other and coming back the same way. Find his average speed.

(a) 2.67 ms^{-1} (b) 2.07 ms^{-1}
(c) 3.67 ms^{-1} (d) 2.60 ms^{-1}

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) :

Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

Average speed = $\frac{160}{60} = \frac{8}{3} = 2.67\,m/s = 2.67\ ms^{-1}$

133. Eddie travels in 9 hours 1 km less than the distance travelled by Fred in 7 hours. In 5 hours Eddie covers 2 km less than the distance travelled by Fred in 4 hours. What is Eddie average speed in km/h?

(a) 10 (b) 12
(c) 9 (d) 11

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (a) Let the speed of Eddie $= V_1$
Speed of fred $= V_2$
According to question,-
$7V_2 - 9V_1 = 1$ ____(i)
$4V_2 - 5V_1 = 2$ ____ (ii)
equation (i) × 4 – equation (ii) × 7
$28V_2 - 36V_1 = 4$
$-28V_2 + 35V_1 = -14$
$-V_1 = -10$
$V_1 = 10$ Km./hr.
Average speed (V_1) of Eddie = 10Km./hr.

134. An Athlete runs a distance of 500 m in 25s. What is the average speed of an athlete?
(a) 20 ms^{-2} (b) 20 ms^{-1}
(c) 20 ms^{1} (d) 20 ms^{2}
RRB Group-D – 26/09/2018 (Shift-III)

Ans : (b) Distance = 500 m, Time = 25s
$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$
$$\text{Speed} = \frac{500}{25} = 20 \text{ m/sec} = 20 \text{ ms}^{-1}$$

135. An object travels 25 m in 3s and then 15 m in 2s. What is the average speed of the object?
(a) 6.67 s^{-1} (b) 6.67 ms^{-1}
(c) 8.0 ms^{-1} (d) 8.0 m
RRB Group-D – 26/09/2018 (Shift-III)

Ans : (c)
$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$
$$= \frac{25+15}{3+2} = \frac{40}{5} = 8\text{m/s} = 8\text{ms}^{-1}$$

136. The object travels 14 m in 4s and then another 16 m in 2s. What is the average speed of the object?
(a) 6.17 ms^{-1} (b) 5.0 s^{-1}
(c) 5.0 ms^{-1} (d) 5.0 m
RRB Group-D – 27/09/2018 (Shift-III)

Ans : (c) Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$
$$\because \quad \text{Average speed} = \frac{(14+16)\text{m}}{(4+2)\text{s}}$$
$$= \frac{30}{6}\text{m/s} = 5 \text{ m/s}$$
Average speed = 5ms^{-1}

137. An object travels 18 m in 6s and then 18 m in 4s. What is the average speed of the object?
(a) 2.1 ms^{-1} (b) 2.6 ms^{-1}
(c) 3.6 ms^{-1} (d) 3.1 ms^{-1}
RRB Group-D – 27/09/2018 (Shift-III)

Ans : (c) Average speed $= \frac{18\text{m}+18\text{m}}{6\text{s}+4\text{s}}$
$$= \frac{36}{10}\text{m/s} = 3.6 \text{ ms}^{-1}$$

138. A person reaches his office in 2 hours from his home. He travels 1/4th of the total distance at a speed of 15 km/h by cycling, 1/2 of the total distance at a speed of 30 km/h by bus and the rest by walking at a speed of 5 km/h. What is the average speed of this person?
(a) 15 Km./hr. (b) 12 Km./hr.
(c) 10 Km./hr. (d) 14 Km./hr.
RRB Group-D – 10/10/2018 (Shift-I)

Ans : (b) Let the total distance = x km
From question–
$$\frac{x}{4\times15} + \frac{x}{2\times30} + \frac{x}{4\times5} = 2$$
$$\frac{x+x+3x}{60} = 2$$
5x = 120
x = 24 km
$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$
$$= \frac{24\text{km}}{2\text{h}} = 12\text{km/h}$$

139. A cyclist covers the first 3 kilometers at an average speed of 8 km/h, the other 2 km at 3 km/h and the last 2 km at 2 km/h. What is the average speed of the entire journey (in km/h)?
(a) 2.4 (b) 3.43
(c) 3.8 (d) 3
RRB Group-D – 06/12/2018 (Shift-II)

Ans. (b) :
$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$
$$= \frac{7}{\frac{3}{8}+\frac{2}{3}+\frac{2}{2}} = \frac{7\times24}{49}$$
$$= \frac{168}{49} = 3.428 \text{ Km./hr.}$$
= 3.43 Km./hr.

140. An object covers a distance of 24 meters in the first 6 seconds and a distance of 16 meters in other 4 seconds. What is the average speed of the object?
(a) 4ms^{-1} (b) 6ms^{-1}
(c) 8ms^{-1} (d) 10ms^{-1}
RRB Group-D – 05/12/2018 (Shift-II)

Ans. (a) Let the total covered distance by object
(S) = 24 + 16 = 40 m.
and total time taken
(t) = 6 + 4 = 10 sec.
Average speed (V) of object = ?
$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$
$$= \frac{40}{10} = 4 \text{ m.s.}^{-1}$$
Hence the speed of object is 4 m.s.$^{-1}$

141. Aditi runs on a circular path of 800 meters and reaches the starting point in 50 seconds. Calculate the average velocity.

(a) $0\ ms^{-1}$ (b) $8.5\ ms^{-1}$

(c) $16\ ms^{-1}$ (d) $10ms^{-1}$

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (a)

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

$$= \frac{\text{Displacement in one round of circumference of the circle}}{\text{Time}}$$

$$= \frac{0}{50} = 0ms^{-1}$$

142. An object covers a distance of 25 meters in 8 seconds and 29 meters in 10 seconds. What is the average speed of the object?

(a) $3\ ms^{-1}$ (b) $3ms^{1}$

(c) $4\ ms^{1}$ (d) $4ms^{-1}$

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (a)

$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

$$= \frac{(25+29)\text{m}}{(10+8)\text{sec}} = \frac{54\text{m}}{18\text{sec}} = 3\ \text{m/sec}$$

143. The odometer of a car shows 3000 km at the beginning of the journey and 3400 km at the end of the journey. If the journey takes 8 hours. Then find the average speed of car (in m/s).

(a) $13.9\ ms^{-1}$ (b) $15.9\ ms^{-1}$

(c) $14.9\ ms^{-1}$ (d) 12.9 ms

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (a) : Total distance covered by Car

= 3400 – 3000 = 400 km

Time taken in travelling = 8 hours

$$\text{Average speed} = \frac{\text{Total Covered Distance}}{\text{Total Time}}$$

$$= \frac{400}{8} = 50\text{km/h}$$

$$= 50 \times \frac{5}{18}\text{m/s}$$

$$= \frac{250}{18} = 13.9\text{m/s}$$

144. The Odometer of a bus shows the reading of 3000 km at the beginning of a journey and 3600 km at the end of the journey. If the journey took a total of 8 hours, then what is the average speed of the bus?

(a) $75kmh^{-2}$ (b) $75kmh^{2}$

(c) $75kmh^{-1}$ (d) $75kmh^{1}$

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (c) Total distance travelled by bus

= 3600–3000 = 600 km

$$\therefore \text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

$$= \frac{600}{8} = 75\ \text{Km./hr.}$$

145. An object travels 20m in 6s and then another 30m in 4s. What is the average speed of the object?

(a) $5\ ms^{1}$ (b) $6\ ms^{-1}$

(c) $5\ ms^{-1}$ (d) $6\ ms^{1}$

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (c)

$$\text{Average speed (V)} = \frac{\text{Total Covered Distance (D)}}{\text{Total Time (t)}}$$

Total distance (D) = $D_1 + D_2$ = 20+30 = 50 m

Total time (t) = $t_1 + t_2$ = 6+4 = 10 Sec.

$$v = \frac{D}{t}$$

$$v = \frac{50}{10}$$

v = 5 m/s

146. The total distance between the two stations is 390 km, a train completes 182 km of this distance at a speed of 56 km/h and 108 km at a speed of 72 km/h. The remaining distance is completed by the train in 1¼ hours. Find the average speed of the train during the entire journey.

(a) 65 Km./hr. (b) 75 Km./hr.

(c) 60 Km./hr. (d) 70 Km./hr.

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (a) : Total distance = 390 km

First distance travelled by train = 182 km

Speed = 56 Km./hr.

Second distance travelled by train = 108 km

Speed = 72 Km./hr.

Remaining Distance = 390 – (182+108) = 100 km

Time $= \frac{5}{4}$ hours

Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{390}{\frac{182}{56}+\frac{108}{72}+\frac{5}{4}} = \frac{390}{\frac{13}{4}+\frac{6}{4}+\frac{5}{4}}$$

$$= \frac{390\times 4}{13+6+5} = \frac{390\times 4}{24}$$

$= 65\,\text{km/hr}$

147. P moves at a speed of 50 km/h in the first 1 hour and at a speed of 70 km/h in the next two hours. What is the average speed of P?

(a) 60 km/hr (b) 63.33 km/hr.

(c) 59.33 km/hr. (d) 62 km/hr.

RRB NTPC 02.04.2016 Shift : 1

Ans : (b) Distance covered by P in first hours = 50 km.

Distance covered by P in second hour = 70

Distance covered by P in third hour = 70 km.

$\therefore$ Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{50+70+70}{1+1+1} = \frac{190}{3} = 63.33 \text{ Km./hr.}$$

148. If P travels the first half of his journey at 40 km/h and the remaining distance at 50 km/h, then what will be the average speed of his total journey?

(a) 44.44 Km./hr. (b) 53.33 Km./hr.

(c) 45 Km./hr. (d) 60 Km./hr.

RRB NTPC 02.04.2016 Shift : 2

Ans : (a) $S_1 = 40$ Km./hr., $S_2 = 50$ Km./hr.

Average speed $= \frac{2\times S_1\times S_2}{S_1+S_2}$

$$= \frac{2\times 40\times 50}{40+50} = \frac{2\times 40\times 50}{90} = \frac{400}{9}$$

= 44.44 Km./hr.

149. A car completes the first half of the distance at a speed of 50 km/h and the remaining half of the journey at 60 km/h. What is the average speed of the car during the entire journey?

(a) 54.54 Km./hr. (b) 36.36 Km./hr.

(c) 50.5 Km./hr. (d) 45.45 Km./hr.

RRB NTPC 02.04.2016 Shift : 3

Ans : (a) Average speed of car $= \frac{2V_1V_2}{V_1+V_2}$

$$= \frac{2\times 50\times 60}{50+60}$$

$$\frac{6000}{110} = 54.54 \text{ Km./hr.}$$

150. A cyclist travels a distance of 8 km at the speed of 15 km/h and a distance of 4 km at a speed of 20 km/h. Then what will be his average speed?

(a) 16.8 (b) 16.36

(c) 15.71 (d) 17.50

RRB NTPC 28.03.2016 Shift : 2

Ans :(b) If a person walks A km distance at a speed of x km./hr. and B km. distance at a speed of y km./hr. then

Average speed $= \frac{A+B}{\frac{A}{x}+\frac{B}{y}} = \frac{8+4}{\frac{8}{15}+\frac{4}{20}} = \frac{12}{\frac{8}{15}+\frac{1}{5}}$

$$= \frac{12}{\frac{8+3}{15}} = \frac{12\times 15}{11} = 16.36\,\text{km/hr}$$

151. A person covers the first 176 km at a speed of 16 km/h and the next 64 km at a speed of 32 km/h. What would be the approximate average speed for the first 240 km of the journey?

(a) 13 km./hr. (b) 27 km./hr.

(c) 18.5 km./hr. (d) 21 km./hr.

RRB NTPC 19.04.2016 Shift : 3

Ans : (c) Time taken to travelled 176 km. distance = $\frac{176}{16}$ = 11 hours

Time taken to travelled 64 km. distance = $\frac{64}{32}$ = 2 hrs.

Average speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

$$= \frac{176+64}{11+2} = \frac{240}{13}$$

= 18.5 Km./hr. (almost)

152. A car travels 30 km at speed of 60 km/h at the begining, the next 20 km travels at a speed of 80 km/h, find its average speed.

(a) 65.67 Km./hr. (b) 65.33 Km./hr.

(c) 66.33 Km./hr. (d) 66.67 Km./hr.

RRB NTPC 06.04.2016 Shift : 1

Ans : (d)

Average speed of car $= \frac{\text{Total Distance}}{\text{Total Time}}$

$= \frac{30+20}{\frac{30}{60}+\frac{20}{80}}$ $\left(\because \text{Time} = \frac{\text{distance}}{\text{speed}}\right)$

$= \frac{50}{\frac{1}{2}+\frac{1}{4}} = \frac{50}{\frac{2+1}{4}} = \frac{50\times 4}{3}$

$= \frac{200}{3} = 66.66 \approx 66.67(\text{km./hr.})$

153. Car P travels a certain distance in 11 hours at a speed of 66 km/h. Car Q travels a distance of 242 km more than Car P in the same time. Find the average speed of Car Q.

(a) 718 Km./hr. (b) 77 Km./hr.
(c) 88 Km./hr. (d) 83 Km./hr.

RRB NTPC 19.01.2017 Shift : 2

Ans : (c) Distance covered by P = 66×11 = 726 km

Average speed of Q $= \frac{\text{Total Distance}}{\text{Total Time}}$

Average speed of Q $= \frac{726+242}{11} = \frac{968}{11} = 88$

Average speed of Q = 88 Km./hr.

154. Arjun travelled a distance of 1000 km from pune to Goa at a speed of 4 km/h and the return journey at a speed of 3 km/h. What was his average speed during the entire journey?

(a) 3 Km./hr. (b) 3.43 Km./hr.
(c) 3.5 Km./hr. (d) 5.4 Km./hr.

RRB NTPC 28.04.2016 Shift : 2

Ans : (b) Average speed $= \frac{2V_1V_2}{V_1+V_2}$

$= \frac{2\times 4\times 3}{(4+3)}$

$= \frac{24}{7} = 3.43$ Km./hr.

Type - 5

155. A certain distance is travelled at a certain speed. If half of the distance is travelled in double time. Find the ratio of both the moves.

(a) 2 : 1 (b) 1 : 3
(c) 4 : 1 (d) 3 : 2

RRB JE - 23/05/2019 (Shift-I)

Ans : (c) Let the certain distance = x km

According to question,

Time = t

Speed = V_1

$V_1 = \frac{x}{t}$ km/h.....(i)

According to second condition–

Distance = x/2 km

Time = 2t

Speed $V_2 = \frac{x/2}{2t} = \frac{x}{4t}$

$\frac{V_1}{V_2} = \frac{\frac{x}{t}}{\frac{x}{4t}} = \frac{4}{1}$

$\boxed{V_1 : V_2 = 4:1}$

156. The ratio of the speeds of two persons while travelling a certain distance is 18:12. Find the ratio of the time taken by them to travel the distance.

(a) 2: 1 (b) 2:5
(c) 2:3 (d) 3:

RRB JE - 23/05/2019 (Shift-III)

Ans : (c) Given-

Ratio of speed = 18 : 12

Ratio of time = $\frac{1}{18}:\frac{1}{12} = \frac{2}{36}:\frac{3}{36} = 2:3$

157. A motor cycle driver covers a distance of 192 km at a speed of 32 km/h. A car starts running from the same place after 2.5 hours the motorcyclist moves,but travels that distance 0.5 hours before. What is the ratio of the speed of the car and the motorcycle?

(a) 3:1 (b) 4:3
(c) 2:1 (d) 1:2

RRB JE - 28/05/2019 (Shift-I)

Ans : (c)

Time taken to cover the distance of 192 km by motorcycle $= \frac{192}{32} = 6\text{h}$

and time taken by car = 6 – (2.5 + .5) = 3 h

Hence Speed of car $= \frac{192}{3} = 64\text{km/hr}$

Hence speed of car : speed of motor cycle

= 64 : 32 = 2 : 1

158. Two motorists traveling in the opposite direction meet at some point in the middle. After this they take 9 and 16 hours respectively to reach their destination. What is the ratio of their speed?

(a) 4:7 (b) 4:3

(c) 5:3 (d) 5:4

RRB JE - 28/05/2019 (Shift-I)

Ans : (b) Let the speed is x and y

Ratio of speed $= \sqrt{\frac{t_2}{t_1}} = \frac{x}{y}$

$\frac{x}{y} = \sqrt{\frac{16}{9}}$

$\frac{x}{y} = \frac{4}{3}$

$x : y = 4 : 3$

159. Mary takes 30 minutes to make one round of a rectangular garden Sila takes 45 minutes to make one round of the same garden. At 9 am both of them started walking in the same direction from the same point. What time will the both meet again at the starting point?

(a) 10 : 45 am (b) 10 : 00 am

(c) 11 : 00 am (d) 10 : 30 am

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (d) Time taken to make a round of the rectangular garden by mary = 30 minutes.

Time taken to make a round of the rectangular garden by Sila = 45 minute

Time to start walking = 9 : 00 AM

meeting time at the starting point

= L.C.M. of 9 : 00 AM + 30 and 45

= 9 : 00 AM + 90 minutes

$= 9 : 00 \text{ AM} + \frac{90}{60} \text{ hours}$

= 10 : 30 AM

160. Car I travels a certain distance in a certain time. Car II travels half of the distance in double the time. Find the ratio of their relative speeds.

(a) 1 : 2 (b) 1 : 4

(c) 2 : 1 (d) 4 : 1

RRB NTPC 17.01.2017 Shift-2

Ans : (d) Let the distance is d and time is t.

According to question,

Speed for car I $(S_1) = \frac{d}{t}$

Speed for car II $(S_2) = \frac{d/2}{2t} = \frac{d}{4t}$

$\therefore S_1 : S_2 = \frac{d}{t} : \frac{d}{4t} = 1 : \frac{1}{4} = 4 : 1$

161. A bus covers some distance at a certain speed. If a person covers one-third of that distance in thrice the time, then the speed of the person is the ratio of the speed of the bus.

(a) 1:3 (b) 3:1

(c) 1:9 (d) 9:1

RRB RPF SI – 13/01/2019 (Shift-II)

Ans : (c) Let the distance d is covered in time t by bus.

speed of bus $= \frac{d}{t}$

from question–

Speed of person $= \frac{d/3}{3t} = \frac{d}{9t}$

Required Ratio $= \frac{d}{9t} : \frac{d}{t} = 1 : 9$

162. Two stations Mumbai and Pune which have a distance of 300 km. Two buses run opposite directions from Mumbai and Pune respectively and cross each other at a distance of 220 km from a station. What is the ratio of their speeds?

(a) 13:9 (b) 10:3

(c) 11:4 (d) 14:5

RRB NTPC 12.04.2016 Shift : 2

Ans : (c) Total distance = 300 km

Distance from first station = 220 km

Distance from second station = 300 – 220 = 80 km

Ratio of speed = 220 : 80 = 11 : 4

163. The ratio of the speeds of two persons while travelling a certain distance is 24:30. Find the ratio of the time taken by them to travel the distance.

(a) 2 : 1 (b) 3:5

(c) 5 : 4 (d) 3: 4

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Ans : (c) Given-

Ratio of speed = 24 : 30

Ratio of time $= \frac{1}{24} : \frac{1}{30} = 5 : 4$